

PROFESSIONAL NURSING COACH

APPLIED MICROBIOLOGY & INFECTION CONTROL

PNC ULTRA-LMR HYBRID MASTER COMPENDIUM • NUR-07 • VERIFIED

Sections A+B • Batches 1–9 • 147 LMR pages + cover & TOC • NORCET / RRB / ESIC

Batch	Units	Pp
B1	Sec-A Units I-II	18
B2	Sec-A Unit III	18
B3	Sec-A Unit IV	18
B4	Sec-B Units I-II	18
B5	Sec-B Units III-V	18
B6	Sec-B Units VI-VII	18
B7	Sec-B Unit VIII	11
B8	Sec-B Units IX-X	12
B9	Sec-B Unit XI final	16

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Pp	Block	Focus
4-21	MIC-B1	Sec-A I-II
22-39	MIC-B2	Sec-A III
40-57	MIC-B3	Sec-A IV
58-75	MIC-B4	Sec-B I-II
76-93	MIC-B5	Sec-B III-V
94-111	MIC-B6	Sec-B VI-VII
112-122	MIC-B7	Sec-B VIII
123-134	MIC-B8	Sec-B IX-X
135-150	MIC-B9	Sec-B XI final

How to use: revise batch-wise with mocks; log errors. Double-QA pending sign-off.

Q	Standard	Status
Q1	All batches merged in unit order	Done
Q2	Classic batches preserved; new batches Hybrid-C	Done
Q3	Footers + TOC ranges verified	Done

INTRODUCTION: WHY MICROBIOLOGY RULES NURSING

Professional Nursing Coach • Relevance to bedside infection control

B+

RAPID REVISION

INC MICR 201 • 3 HRS

Core: Microbiology = study of microbes invisible to naked eye. Nurse relevance = break chain of infection, safe specimen, correct sterilization, HAI prevention.

IMPORTANCE & RELEVANCE TO NURSING

- **Infection chain breaker:** nurse controls reservoir → portal of exit → transmission → portal of entry → susceptible host.
- **Bedside links:** hand hygiene, PPE, BMW segregation, specimen timing (before antibiotics), vaccine cold chain, outbreak reporting.
- **HAI bundle ownership:** CAUTI, CLABSI, SSI, VAP prevention is nursing audit cue in NORCET.
- **Antimicrobial stewardship:** collect culture FIRST, then start antibiotics; review 48-72 hrs.

Term	Exam meaning	Nursing cue
Sterilization	Destroys ALL microbes incl. spores	Autoclave for critical items
Disinfection	Destroys vegetative pathogens, not all spores	Semi-critical items, surfaces
Antiseptic	Safe on living tissue	Skin prep, hand rub
Asepsis	Freedom from infection / technique	Surgical vs medical asepsis

FATAL TRAP: Calling disinfection = sterilization. Disinfected endoscope is NOT spore-free — never use for critical sterile cavity entry.

NORCET HIT: Mnemonic **S-D-A**: Sterilize things, Disinfect surfaces, Antiseptic on skin.

Speed Check: FIRST nursing action before starting antibiotics in suspected sepsis?

Collect blood cultures (2 sets) FIRST, then start broad-spectrum antibiotics — never reverse.

HISTORY & CONCEPTS IN 5 MINUTES

Fathers, postulates, principles every NORCET MCQ repeats

B+

TIMELINE CARD

HISTORY + TERMS

Core: Leeuwenhoek saw → Pasteur proved germs → Koch linked germ-disease → Lister antiseptis → Fleming penicillin.

Scientist	Contribution	Trap word
Antonie van Leeuwenhoek	Father of microbiology; first microscope observation of bacteria	MICROSCOPE
Louis Pasteur	Germ theory; pasteurization; rabies/anthrax vaccines	PASTEURIZATION
Robert Koch	Koch postulates; TB bacillus; cholera vibrio; solid media	POSTULATES
Joseph Lister	Father of antiseptic surgery (carbolic acid)	ANTISEPSIS
Alexander Fleming	Penicillin from <i>Penicillium notatum</i> (1928)	PENICILLIN
Christian Gram (1884)	Gram stain differential	GRAM +/-

KOCH POSTULATES — 4 STEPS

- 1. Organism found in ALL diseased, absent in healthy. 2. Isolated in pure culture. 3. Inoculated culture reproduces disease in healthy host. 4. Re-isolated same organism.
- Exception trap:** viruses, leprosy bacillus (cannot grow on artificial media) — postulates NOT fulfilled, still pathogens.

FATAL TRAP: Pasteur ≠ Koch. Pasteur = germ theory/vaccine; Koch = postulates + pure culture. MCQ swaps them.

NORCET HIT: LL-P-K-L-F-G: Leeuwenhoek Lens, Pasteur Germ, Koch Cause, Lister Clean, Fleming Fungus-drug, Gram Grade.

Speed Check: Who is Father of antiseptic surgery?

Joseph Lister — carbolic acid spray for surgical wounds.

BACTERIAL CELL: STRUCTURE MASTER

What each layer does + why antibiotics target it

A+

DIAGRAM + TABLE

STRUCTURE

Core: Bacteria = prokaryote (no true nucleus, no mitochondria). Cell wall gives shape + Gram reaction; capsule resists phagocytosis; spore resists sterilization.

Structure	Function	Clinical link
Capsule / slime layer	Anti-phagocytic, adhesion	Pneumococcus, Klebsiella, Meningococcus virulent; negative stain
Cell wall (peptidoglycan)	Shape, osmotic protection	Gram+ thick, Gram- thin + LPS endotoxin; penicillin target
Cell membrane	Selective permeability	Polymyxin target
Mesosome	Respiration analogue	Bacterial mitochondrion equivalent
Ribosome 70S	Protein synthesis (50S+30S)	Erythromycin (50S), streptomycin (30S); human 80S — selective toxicity
Plasmid	Extra-chromosomal DNA — resistance	AMR spread by conjugation
Flagella / Pili / Fimbriae	Motility / conjugation / adhesion	E. coli UTI adhesion; motility in hanging drop
Spore (endo)	Survival in heat/chemicals	Bacillus + Clostridium only; autoclave needed

FATAL TRAP: Mycoplasma has NO cell wall — Gram stain useless, penicillin useless, fried-egg colonies.

NORCET HIT: 70S = Bacteria, 80S = Human. Drug hits 70S, spares you.

Speed Check: Which structure carries drug-resistance genes?

Plasmid — transferred by conjugation (R-factor).

MORPHOLOGY: COCCI VS BACILLI SHAPES

Arrangement tells organism before stain

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VISUAL MATRIX

SIZE + SHAPE

Core: Cocci = round; Bacilli = rod; Spirilla = spiral. Arrangement + Gram reaction = first NORCET clue. Size: bacteria 0.5–5 µm; viruses 20–300 nm.

Shape / Arrangement	Classic organism clue	Disease
Gram+ cocci in clusters	Staphylococcus	Boil, MRSA, food poisoning
Gram+ cocci in chains	Streptococcus	Pharyngitis, rheumatic fever
Gram+ cocci in pairs (lancet)	Pneumococcus (Diplococcus)	Pneumonia, meningitis
Gram- cocci kidney-bean pairs	Neisseria (gonorrhoeae/meningitidis)	Gonorrhoea / meningitis
Gram+ bacilli with spores	Clostridium (anaerobic) / Bacillus (aerobic)	Tetanus, gas gangrene / Anthrax
Gram- bacilli (non-sporing)	E. coli, Klebsiella, Pseudomonas	UTI, HAI, pneumonia
Comma / curved	Vibrio cholerae	Rice-water diarrhoea
Spiral / corkscrew	Treponema, Leptospira	Syphilis / Weil disease

SIZE: Bacteria µm

Virus nm (1000x smaller)

MOTILITY: Hanging drop

FATAL TRAP: All bacilli are NOT spore-formers. Only Bacillus + Clostridium make spores. E. coli never makes spores.**NORCET HIT:** Clusters = Staph, Chains = Strep, Pairs = Pneumo/Neisseria.**Speed Check: Kidney-bean diplococci on Gram stain = ?**

Neisseria — intracellular Gram-negative diplococci.

GROWTH, NUTRITION & ENVIRONMENT

Temp, moisture, blood — what pathogens demand

A+

CONCEPT + TABLE

GROWTH NEEDS

Core: Most pathogens = mesophiles growing best at 37°C, moist, blood/body-fluid rich, pH 7.2–7.6. Dry + cold slows; autoclave kills.

Factor	Types	Nursing / Lab cue
Temperature	Psychrophile (cold), Mesophile 20–45°C, Thermophile (heat)	Incubate cultures at 37°C; fridge 2–8°C only preserves, not kills
Oxygen	Obligate aerobe (TB), anaerobe (Clostridium), facultative (E.coli), microaerophilic (Campylobacter)	Anaerobic jar/GasPak for wound deep cultures
Moisture + nutrition	Need water, carbon, nitrogen; blood-enriched for fastidious	Dry swab kills Neisseria — use transport media fast
pH / CO ₂	Neutral 7.2–7.6; Neisseria needs 5–10% CO ₂	Candle jar for pneumococcus
Growth curve	Lag → Log (exponential) → Stationary → Decline	Log phase = antibiotic most effective
Colonization vs infection	Colonization = present, no damage; Infection = invasion + response	Do NOT treat mere colonization (except MRSA screen protocol)

FATAL TRAP: Refrigeration does NOT sterilize. It only arrests growth — rewarming revives pathogens.**NORCET HIT: 37-7.4-moist:** Human temp, human pH, human fluids = pathogen paradise.**Speed Check: Best temperature to incubate most pathogenic cultures?**

37°C (mesophilic range) — human body temperature.

CLASSIFICATION OF MICROBES AT A GLANCE

Bacteria, virus, fungus, parasite — one table

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COMPARE TABLE

HIGH-YIELD

Core: Prokaryote (bacteria) vs Eukaryote (fungi/parasites) vs Acellular (viruses). Cell wall present except virus + Mycoplasma.

Microbe	Cell type / Size	Key clue	Example
Bacteria	Prokaryote, 0.5–5 µm	Gram stain + culture on agar	Staph, E.coli, TB
Virus	Acellular, 20–300 nm; DNA/RNA only	Obligate intracellular; cannot grow on agar	HIV, Hepatitis, Rabies
Fungus	Eukaryote; yeast/hyphae	KOH + Sabouraud; 25°C + 37°C dimorphic	Candida, Aspergillus
Parasite	Eukaryote; protozoa/helminth	Stool ova/parasite; vector-borne	Plasmodium, Ascaris
Rickettsia / Chlamydia	Obligate intracellular bacteria	Cannot grow on cell-free media; Weil-Felix	Typhus, trachoma
Mycoplasma	No cell wall, smallest free-living	Fried-egg colonies; resistant to penicillin	Atypical pneumonia

FATAL TRAP: Prescribing penicillin for virus/atypical pneumonia. Penicillin needs cell wall — viruses + Mycoplasma have none.**NORCET HIT: B-V-F-P:** Bacteria (agar), Virus (cell), Fungus (KOH), Parasite (stool).**Speed Check: Smallest free-living organism without cell wall?**

Mycoplasma — pleomorphic, penicillin-resistant.

MOTILITY, COLONIZATION & NORMAL FLORA

When movement + resident flora matters

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CONCEPT CARD

FLORA vs PATHOGEN

Core: Motility by flagella (hanging drop). Normal flora protects; disruption causes opportunistic infection (e.g. Candida after broad antibiotics).

- **Motile:** E.coli, Vibrio, Proteus (swarming), Salmonella (except Typhi galli), Pseudomonas. **Non-motile:** Staph, Strep, Pneumo, Klebsiella, Shigella, Neisseria.
- **Hanging drop:** live wet mount for darting motility (Vibrio) — do NOT confuse with Brownian movement.
- **Normal flora sites:** skin (Staph epidermidis), nasopharynx, gut (E.coli, Lactobacillus), vagina (Lactobacillus/Doderlein).
- **Sterile sites:** blood, CSF, pleural fluid, urine in bladder, lower airway — any growth = significant.
- **Colonization trap:** nasal MRSA carrier, catheter colonization without fever — follow protocol, avoid blind antibiotics.

Site	Resident flora	Sterile?
Skin / nares	Staph epidermidis, Corynebacterium	No
Blood / CSF	None — sterile	YES — culture positive = infection
Gut / vagina	E.coli, Lactobacillus	No — protective

FATAL TRAP: Treating urine culture with mixed flora + no symptoms as UTI. Likely contamination — repeat midstream clean-catch.

NORCET HIT: Sterile = Blood, CSF, Bladder urine, Pleura. Growth there is never normal.

Speed Check: Test for live bacterial motility?

Hanging drop preparation — true directional motility.

GRAM STAIN: THE #1 NORCET STAIN

Purple vs Pink decides antibiotic in seconds

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PROCEDURE + TRAP

1884 CHRISTIAN GRAM

Core: Gram+ = thick peptidoglycan retains crystal violet (purple). Gram- = thin + LPS, decolorized, takes safranin (pink).

Step	Reagent	What happens
1. Primary stain	Crystal violet (1 min)	All cells purple
2. Mordant	Gram iodine (1 min)	CV-iodine complex fixed
3. Decolorizer	Acetone-alcohol (10-20 sec)	Gram- loses purple; Gram+ retains
4. Counterstain	Safranin (30 sec)	Gram- pink/red; Gram+ stays purple

- **Gram+ clue:** purple cocci clusters/chains, bacilli with spores. **Gram- clue:** pink rods, kidney-bean diplococci.
- **Specimen:** pus, sputum, urethral discharge, CSF — heat-fixed smear, thin, even.

FATAL TRAP: Over-decolorizing (>30 sec) turns Gram+ falsely pink. Under-decolorizing leaves Gram- falsely purple. Time it.

NORCET HIT: V-I-A-S: Violet → Iodine → Alcohol → Safranin. Purple = Positive.

Speed Check: Colour of Staph aureus on correct Gram stain?

Purple Gram-positive cocci in clusters.

AFB STAIN: ZIEHL-NEESEN FOR TB

Red bacilli on blue = Mycobacterium

A+**ZN PROCEDURE****MYCOLIC ACID****Core:** Mycolic-acid wall resists acid decolorizer → retains carbol fuchsin (red). Non-AFB takes methylene blue counterstain.

ZN Step	Reagent / Action	Result
Primary (heat)	Carbol fuchsin + steam 5 min	All red
Decolorizer	20% H ₂ SO ₄ (AFB resists) / 3% HCl-alcohol	AFB stays red; others colorless
Counterstain	Methylene blue	Background blue; AFB bright red beaded rods

- **Organisms:** M. tuberculosis, M. leprae, Nocardia (partial), Cryptosporidium oocysts (modified ZN).
- **Specimens:** sputum (spot + morning), gastric aspirate, CSF, lymph node aspirate. Need 10,000 bacilli/mL to be smear-positive.
- **Grading + LED-FM:** Auramine-rhodamine fluorescent microscopy more sensitive for RNTCP/NTEP.

FATAL TRAP: Smear-negative ≠ TB ruled out. Culture (LJ) / CBNAAT needed. Never start/stop ATT on one smear.

NORCET HIT: Red on Blue = TB. 20% acid proof = Myco wall.

Speed Check: Decolorizer in ZN stain?

20% sulphuric acid — AFB resists due to mycolic acid.

CAPSULE, SPORE & SIMPLE STAINS

Negative, Schaeffer-Fulton, Albert — when each

A+

STAIN SELECTOR

CAPSULE + SPORE

Core: Capsule seen by what it excludes (negative stain). Spore seen by forcing dye in with heat (malachite green).

Stain	For / Reagent	Appearance
Simple (methylene blue)	Shape, arrangement	Blue cells on clear field
Negative (India ink / Nigrosin)	Capsule — Strep pneumo, Klebsiella, Cryptococcus	Clear halo around cell on black field
Spore (Schaeffer-Fulton)	Malachite green + steam; safranin counter	Green spore, pink vegetative cell
Albert (metachromatic granules)	C. diphtheriae volutin granules	Green bacilli + bluish-black granules
Gram vs capsule trap	Capsule NOT stained by Gram	Need negative stain to prove capsule

- Spore position clue:** Central (Bacillus anthracis), Terminal drumstick (Clostridium tetani), Subterminal (Clostridium perfringens).

FATAL TRAP: Heat-fixing a capsule smear destroys capsule. Air-dry only for negative stain.

NORCET HIT: Drumstick = Tetani terminal spore. Halo on black = Capsule.

Speed Check: Stain for pneumococcal capsule in CSF?

Negative staining with India ink — clear halo.

KOH MOUNT & LPCB FOR FUNGI

Skin, nail, sputum to hyphae in minutes

A+

FUNGAL CARD

KOH + LPCB

Core: 10% KOH dissolves keratin, leaves refractile hyphae/spores. LPCB stains chitin wall blue for morphology.

Test	Specimen	Finding
10% KOH mount	Skin scraping, nail clipping, vaginal discharge, sputum	Branching septate hyphae (dermatophytes/Aspergillus) or budding yeast (Candida)
LPCB mount	Culture growth from Sabouraud	Blue hyphae + conidia arrangement — species ID
Germ tube	Candida in serum at 37°C, 2 hrs	Germ tube positive = <i>C. albicans</i>
India ink (fungal)	CSF for Cryptococcus	Halo capsule of budding yeast

- **Collection cue:** scrape active edge of lesion, not center; transport dry, not refrigerated for dermatophytes.
- **Safety:** KOH corrosive — wear gloves, avoid eye contact; LPCB contains phenol.

FATAL TRAP: Sending fungal scraping in formalin kills culture. Send dry in sterile paper envelope / petri dish.

NORCET HIT: KOH = bedside hyphae; LPCB = lab morphology; Germ tube = *albicans*.

Speed Check: Bedside test for oral thrush pseudohyphae?

10% KOH mount — budding yeast + pseudohyphae.

STAIN SELECTOR: WHICH TEST WHEN?

One-page decision matrix for MCQs

A+**DECISION TABLE****EXAM TRAP-PROOF****Core:** Pus/urethral = Gram; chronic cough = ZN; capsule/meningitis = India ink; skin/nail = KOH; culture morphology = LPCB.

Clue / Specimen	FIRST stain	What proves
Pus, wound, urethral discharge	Gram	Purple vs pink + arrangement
Sputum, chronic cough, lymph node	ZN (AFB)	Red beaded rods = TB
CSF with capsule suspicion	India ink negative	Halo = Crypto/pneumo capsule
Skin scale / nail / vaginal curd	10% KOH	Hyphae / budding yeast
Suspected diphtheria membrane	Albert	Metachromatic granules
Suspected anthrax / tetanus	Spore stain	Green spore position
Fungal culture on SDA	LPCB	Conidia pattern

FATAL TRAP: Ordering Gram stain to diagnose TB or KOH to diagnose bacteria — wrong test, wasted time, false negative.**NORCET HIT:** Read specimen first, then pick stain. Specimen dictates stain.**Speed Check: Throat pseudomembrane + bull neck — stain?**
Albert stain for *C. diphtheriae* metachromatic granules.

CULTURE MEDIA: BASAL TO SPECIAL

Solid, liquid, defined — what grows where

A+

MEDIA TREE

ENRICHED vs SELECTIVE

Core: Agar solidifies (1.5-2%), broth is liquid. Enriched feeds fastidious; Selective suppresses; Differential colours; Enrichment amplifies.

Media class	Example	Use
Basal / simple	Nutrient agar/broth, peptone water	Non-fastidious growth; motility/indole
Enriched	Blood agar, chocolate agar, Loeffler serum	Fastidious: Strep, Neisseria, Haemophilus
Selective	MacConkey (Gram-neg only), LJ (TB), TCBS (Vibrio)	Suppress unwanted flora
Differential	MacConkey (lactose), Blood agar (hemolysis), CLED	Colour/hemolysis distinguishes
Enrichment (liquid)	Selenite F, tetrathionate, alkaline peptone water	Amplify Salmonella/Shigella/Vibrio from stool
Special	Sabouraud (fungi), LJ/Dorset (TB), Bordet-Gengou	Fungus / mycobacteria / Bordetella

FATAL TRAP: Calling MacConkey only selective. It is BOTH selective (bile + crystal violet) AND differential (lactose + neutral red).**NORCET HIT:** Blood = enriched + differential (hemolysis). MacConkey = selective + differential.**Speed Check: Media showing beta-hemolysis around Strep colonies?**

Blood agar — enriched + differential by hemolysis.

CLASSIC MEDIA–ORGANISM PAIRS

Memorize these 8 for every MCQ

A+**PAIR MATRIX****MUST-MEMORIZE****Core:** Chocolate = Haemophilus; LJ = TB; SDA = fungi; TCBS = Vibrio; XLD/DCA = Salmonella/Shigella.

Media	Organism / Finding	Clue
Chocolate agar	Haemophilus, Neisseria (fastidious)	Heated blood (lysed RBC), needs X+V factors
Lowenstein-Jensen (LJ)	M. tuberculosis — buff, rough colonies in 2-8 wks	Egg-based, malachite green selective
Sabouraud dextrose (SDA)	Fungi — acidic pH 5.6, 25°C	Low pH suppresses bacteria
MacConkey	E.coli pink (LF) vs Salmonella pale (NLF)	Lactose fermenter = pink
TCBS	Vibrio cholerae — yellow colonies	Thiosulphate-citrate-bile-sucrose
Bordet-Gengou / Regan-Lowe	Bordetella pertussis	Potato-blood-glycerol
Robertson cooked meat	Anaerobes (Clostridium)	Meat + paraffin excludes O ₂
CLED	Urine culture — prevents Proteus swarming	Cystine-lactose-electrolyte-deficient

FATAL TRAP: Incubating SDA at 37°C only. Dimorphic fungi need dual temp: 25°C (mould) + 37°C (yeast).**NORCET HIT: C-L-S-T:** Chocolate-Haemo, LJ-TB, SDA-Fungus, TCBS-Vibrio.**Speed Check: Yellow colonies on TCBS = ?**

Vibrio cholerae — sucrose fermenting.

STREAK, POUR, SPREAD & DILUTION

Pure culture techniques step-wise

A+

TECHNIQUE STEPS

PURE CULTURE

Core: Pure culture = single species. Streak dilutes on solid surface; pour mixes in molten agar; spread spreads measured volume.

Technique	How	Counts? / Use
Streak plate (quadrant)	Loop streaks + flaming between quadrants; isolated colonies	No count; isolation + morphology
Pour plate	Sample mixed in 45°C molten agar, poured, incubated	Colony count (CFU/mL); heat-tolerant only
Spread plate	0.1 mL spread with L-rod on preset agar	Viable count; antibiotic sensitivity lawn
Tube dilution / MIC	Serial doubling dilutions + organism	MIC/MBC for antibiotics
Stroke / lawn	Even swab over entire plate	Antibiotic disc diffusion (Kirby-Bauer)

- **Sterile loop cue:** flame to red-hot, cool 10 sec before touching culture — hot loop kills + aerosols.

FATAL TRAP: Flaming loop and streaking immediately. Hot loop kills inoculum → false no-growth. Cool it.

NORCET HIT: Isolation = Streak; Counting = Pour/Spread; Sensitivity = Lawn + discs.

Speed Check: Method for isolated single colonies from mixed pus?

Quadrant streak plate with inter-quadrant flaming.

ANAEROBIC CULTURE & TRANSPORT

GasPak, McIntosh jar, RCM — no oxygen

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ANAEROBE CARD

O₂-FREE**Core:** Deep wound, gas gangrene, intra-abdominal pus need O₂-free culture. Expose to air = kill anaerobe = false negative.

Method / Media	Principle	Nursing cue
McIntosh-Fildes jar	Evacuation + H ₂ + catalyst; indicator methylene blue	Lab gold standard; anaerobe jar
GasPak sachet	H ₂ + CO ₂ generator + palladium catalyst	Bedside jar; water activates sachet
Robertson cooked meat (RCM)	Meat consumes O ₂ ; paraffin seal	Transport + culture for Clostridium
Thioglycollate broth	Reducing agent gradient; growth at bottom = anaerobe	Check growth level to read O ₂ need
Transport rule	Anaerobe swab in Stuart/Amies + fast dispatch (<2 hrs)	Never refrigerate anaerobe specimen; keep room temp, urgent

FATAL TRAP: Refrigerating anaerobic pus or leaving swab overnight on desk. Fastidious anaerobes die — always urgent transport.**NORCET HIT:** Bottom growth in thioglycollate = strict anaerobe. Top = aerobe.**Speed Check: Indicator that anaerobic jar is truly anaerobic?**

Methylene blue strip turns colourless (reduced).

BEFORE CULTURE: COLLECTION RULES

Right specimen, right time, right container

A+**PRE-ANALYTICAL****NURSE-OWNED**

Core: Collect BEFORE antibiotics, from active site, in sterile container, label bedside, transport <2 hrs. Contamination ruins culture.

- **Timing:** blood cultures during fever spike/rigor, 2 sets (aerobic + anaerobic) from 2 venepunctures before antibiotics.
- **Midstream urine:** clean-catch after peri-wash, discard first part, sterile wide-mouth container, transport within 2 hrs or 4°C up to 24 hrs.
- **Sputum:** early morning, deep cough, not saliva; rinse mouth with water only.
- **Wound:** clean with saline, sample leading edge/pus aspirate — swab is last choice.
- **Labelling:** name, ID, site, date-time, antibiotics status, clinical diagnosis — bedside, never lab desk.

FATAL TRAP: Collecting culture AFTER first antibiotic dose then reporting sterile as true negative. Always document post-antibiotic status.

NORCET HIT: B-S-W-U: Before antibiotics, Sterile container, Site-active, Urgent (<2h).

Speed Check: How many blood culture sets before antibiotics?

2 sets from 2 separate venepunctures (aerobic + anaerobic each).

BATCH 1 ONE-LINER BLITZ (18 IN 3 MIN)

Cover Units I-II: intro + morphology + stains + media

A+**SPEED REVISION****18 Q&A****Use:** Cover answers, recall aloud. All from Batch 1 pages 1–17.

#	One-liner Q	A
1	Father of microbiology?	Leeuwenhoek
2	Koch postulates exception?	Viruses / lepra bacillus (no artificial culture)
3	Bacteria size unit?	Micron (0.5–5 µm)
4	No cell wall organism?	Mycoplasma
5	Clusters vs chains?	Staph vs Strep
6	Gram+ colour?	Purple (crystal violet retained)
7	ZN decolorizer?	20% H ₂ SO ₄
8	Capsule stain?	India ink negative
9	Drumstick spore?	C. tetani terminal
10	KOH for?	Fungal hyphae
11	LPCB for?	Fungal morphology from culture
12	MacConkey dual role?	Selective + differential
13	LJ medium for?	TB (egg-based)
14	TCBS yellow?	Vibrio cholerae
15	Isolation technique?	Quadrant streak
16	Anaerobic indicator?	Methylene blue colourless
17	Blood cultures timing?	Before antibiotics, 2 sets
18	Sterile sites?	Blood, CSF, bladder urine, pleura

Next: Batch 2 = Section A Unit III (Cocci, Bacilli, Viruses, Fungi, Parasites) — 5+5+4+3+4 pages.

STAPHYLOCOCCUS AUREUS MASTER

Clusters, coagulase, MRSA, food poisoning

A+

COCCI CARD

GRAM+ CLUSTERS

Core: Gram+ cocci in clusters, catalase +, coagulase + (aureus). Skin/nares carrier; pus, boil, MRSA, TSS, scalded skin, food poisoning.

Feature	Detail	Nursing clue
Morphology	Gram+ clusters (grape-like), non-motile, non-sporing	Pus with clusters = Staph till proved otherwise
Culture	Blood agar beta-hemolysis; Mannitol salt (yellow); DNase +	Yellow on MSA = mannitol fermenter
Coagulase test	Aureus +, epidermidis/saprophyticus -	Coagulase + = aureus (pathogenic)
Toxins	TSST-1, exfoliatin, enterotoxin (heat-stable), PVL	Tampon TSS; Ritter disease; 2-6 hr vomiting after cream food
MRSA	mecA gene, PBP2a; ceftioxin screen; contact precautions	Nasal mupirocin + chlorhexidine bath per protocol

FATAL TRAP: Giving beta-lactam for MRSA. MRSA resists ALL beta-lactams — vancomycin/linezolid per culture, plus isolation.**NORCET HIT:** Clusters + Coagulase + = Aureus. Yellow MSA + Beta lysis.**Speed Check: Test separating *S. aureus* from epidermidis?**

Tube coagulase — aureus positive, epidermidis negative.

STREPTOCOCCUS: PYOGENES & AGALACTIAE

Chains, hemolysis, ASLO, rheumatic fever

A+**CHAINS CARD****BETA vs ALPHA**

Core: Gram+ chains, catalase **NEGATIVE** (vs Staph +). Pyogenes = Group A beta-hemolytic; Pneumoniae = alpha; Agalactiae = Group B neonatal.

Organism	Hemolysis / Test	Disease
S. pyogenes (Group A)	Beta-hemolysis; bacitracin sensitive; PYR +; ASLO rise	Pharyngitis, impetigo, erysipelas, necrotizing fasciitis, RF, PSGN
S. agalactiae (Group B)	Beta; CAMP +; vaginal/rectal screen 35-37 wks	Neonatal sepsis/meningitis; intrapartum penicillin prophylaxis
S. pneumoniae	Alpha (green); optochin sensitive; bile soluble; Quellung +	Lobar pneumonia, meningitis, otitis (see p3)
Viridans / Enterococcus	Alpha/gamma; BEA +; VRE watch	Endocarditis after dental work; UTI

- **Post-strep sequelae:** Rheumatic fever (2-3 wks, Jones criteria, penicillin prophylaxis) vs PSGN (cola urine, hypertension).

FATAL TRAP: Catalase confuses Staph vs Strep. Staph +, Strep -. Never reverse.

NORCET HIT: Chains + Catalase - + Beta = Pyogenes. ASLO = recent Group A.

Speed Check: Prophylaxis for Group B positive mother in labour?

IV penicillin/ampicillin intrapartum to prevent neonatal sepsis.

PNEUMOCOCCUS & NEISSERIA TWINS

Lancet pairs vs kidney-bean pairs

A+

DIPLOCOCCI

PAIRS MATRIX

Core: Both diplococci, both meningitis, opposite Gram + draught sensitivity. Pneumo = Gram+ lancet; Neisseria = Gram- kidney-bean.

Feature	Pneumococcus	Neisseria meningitidis / gonorrhoeae
Gram / shape	Gram+ lancet pairs, capsulated	Gram- kidney-bean intracellular pairs
Culture	Alpha-hemolysis; optochin S; bile soluble; needs CO ₂	Thayer-Martin/chocolate; oxidase +; dies on drying/cold
Disease	Lobar pneumonia (rusty sputum), meningitis, otitis	Meningococemia (purpura) / gonorrhoea (green discharge)
Vaccine / Px	Pneumococcal (elderly, splenectomy); amoxicillin	Meningococcal (crowding); ceftriaxone + droplet precautions

FATAL TRAP: Refrigerating Neisseria swab. Cold + drying kills it — transport warm, immediate, on charcoal media.**NORCET HIT:** Lancet + Optochin = Pneumo. Bean + Oxidase = Neisseria.**Speed Check: Rusty sputum + lobar consolidation = ?**

Pneumococcal pneumonia — Gram+ lancet diplococci.

DIPHTHERIA & LISTERIA ALERTS

Pseudomembrane + bull neck; cold-growing rods

A+

BACILLI I

ALBERT + TUMBLING

Core: C. diphtheriae = Chinese-letter bacilli, metachromatic granules, ELEK toxin test. Listeria = cold growth, tumbling motility, pregnancy danger.

Organism	Lab clue	Disease / Action
C. diphtheriae	Albert granules; Tinsdale black colonies; Loeffler; ELEK toxigenicity	Grey adherent membrane + bull neck; DPT vaccine; DAT antitoxin + penicillin/erythro; droplet + contact
Listeria monocytogenes	Gram+ short rods; tumbling at 25°C; cold enrichment 4°C; BEA +	Soft cheese/unpasteurized milk; abortion, neonatal meningitis; avoid in pregnancy; ampicillin
Nursing red flag	Never try to remove diphtheria membrane (bleeds)	Airway watch, stridor, myocarditis ECG, palatal palsy

FATAL TRAP: Forcefully removing throat membrane. Causes bleeding + toxin release + airway obstruction. Suction ready, no scraping.

NORCET HIT: Bull neck + Membrane = Diphtheria. Cold milk + Abortion = Listeria.

Speed Check: Test proving diphtheria toxin production?

ELEK gel precipitation test for toxigenicity.

CLOSTRIDIUM TRIO: TETANI, PERFRINGENS, BOTULINUM

Anaerobic spore-formers, exotoxins

A+**ANAEROBE TRIO****SPORE + TOXIN****Core:** All Gram+ anaerobic bacilli with spores. Tetani = rigid; Perfringens = gas; Botulinum = floppy.

Organism	Spore / Toxin	Disease / Care
C. tetani	Terminal drumstick; tetanospasmin (blocks GABA/glycine)	Lockjaw, risus sardonicus, opisthotonus; TIG + metronidazole + dark quiet room; TT vaccine
C. perfringens	Subterminal; alpha toxin; stormy fermentation; Nagler +	Gas gangrene (crepitus, foul), food poisoning; debridement + penicillin + hyperbaric
C. botulinum	Subterminal; botulinum (blocks ACh) — most potent	Floppy baby, diplopia, canned food; antitoxin + no honey <1 yr; no aminoglycosides

FATAL TRAP: Stimulating tetanus patient (light/noise). Triggers painful spasms + laryngospasm. Nurse in dark, quiet, minimal handling.**NORCET HIT:** Rigid = Tetani, Gas = Perfringens, Floppy = Botulinum.**Speed Check: Infant food absolutely avoided to prevent botulism?**

Honey before 1 year — C. botulinum spores.

ENTEROBACTERALES: E. COLI, KLEBSIELLA, PROTEUS

UTI triad + lactose + swarming

A+**GRAM- RODS****UTI + HAI****Core:** Gram- non-sporing rods, facultative. E.coli = #1 UTI; Klebsiella = capsule + currant jelly; Proteus = swarming + struvite stones.

Organism	Culture / Test	Disease
E. coli	MacConkey pink (LF); EMB metallic sheen; indole +	UTI, traveller diarrhoea, neonatal meningitis, HUS (O157)
Klebsiella pneumoniae	Mucoid capsule; non-motile; urease slow; currant-jelly sputum	Lobar pneumonia (alcoholics), UTI, CAUTI, VAP
Proteus mirabilis	Swarming on blood agar; urease +++; Dienes line	Struvite staghorn stones (alkaline urine), UTI with catheter
Pseudomonas aeruginosa	Blue-green pyocyanin; fruity/grape odour; oxidase +; non-fermenter	Burn wound, VAP, green discharge; ceftazidime + contact

FATAL TRAP: Reporting Proteus swarming as mixed growth. It swarms and covers all — use CLED to inhibit swarming for urine.**NORCET HIT:** Pink = E.coli, Mucoid = Klebsiella, Swarming = Proteus, Green = Pseudo.**Speed Check: Staghorn stone + alkaline urine + UTI = ?**

Proteus — urease splits urea → ammonia → struvite.

SALMONELLA, SHIGELLA, CHOLERA

Enteric fever vs dysentery vs rice-water

A+

ENTERIC FEVER CARD

WIDAL + STOOL

Core: Salmonella = motile (except galli), H₂S +; Shigella = non-motile, no H₂S; Vibrio = comma, oxidase +.

Organism	Key test	Disease / Care
S. Typhi	Blood culture week 1; Widal week 2 (O+H); XLD black centre	Step-ladder fever, rose spots, bradycardia; ceftriaxone; carrier in gallbladder
Shigella	Non-motile, NLF, no H ₂ S; RBC + pus, no RBC casts	Bacillary dysentery (small frequent stools with blood-mucus, tenesmus); ORS + fluids
V. cholerae	TCBS yellow; darting motility; oxidase +	Rice-water stools, severe dehydration; ORS + doxycycline; ORS corner
Campylobacter / H. pylori	Microaerophilic; urea breath +	Guillain-Barre trigger / ulcer — triple therapy

FATAL TRAP: Doing Widal in week 1 to rule out typhoid. Antibodies appear week 2 — week 1 needs blood culture.**NORCET HIT:** Week 1 Blood, Week 2 Widal, Week 3 Stool for Typhi.**Speed Check: Rice-water stools + severe dehydration = ?**

Cholera — Vibrio cholerae, TCBS yellow, ORS stat.

MYCOBACTERIA & ZOONOTIC BACILLI

TB, leprosy, anthrax, plague at a glance

A+**AFB + ZOONOSIS****LJ + SAFETY****Core:** TB = aerobic, LJ slow; Leprae = cannot culture, armadillo/mouse footpad; Anthrax = medusa-head, McFadyean reaction.

Organism	Clue	Disease / precaution
M. tuberculosis	AFB red, LJ buff 2-8 wks, CBNAAT +	Cavitary U/L lobe, night sweats; airborne, N95, ATT DOTS
M. leprae	Cannot grow on LJ; slit-skin smear; BI/MI	Hypopigmented anesthetic patch; MDT (rifa+dapsone+clofa); no isolation needed on treatment
B. anthracis	Medusa-head; McFadyean polychrome; non-motile	Malignant pustule, woolsorter pneumonia; ciprofloxacin + vaccine
Yersinia / Brucella	Bipolar safety-pin; milk-borne	Plague (bubonic) / undulant fever — barrier + doxycycline

FATAL TRAP: Isolating leprosy patient long-term. Once on MDT, non-infectious — isolation is stigma, not science.**NORCET HIT:** Medusa = Anthrax, Safety-pin = Plague, No culture = Leprae.**Speed Check: Culture proving TB when smear negative?**

LJ egg medium — gold standard, 2-8 weeks.

VIRUSES: STRUCTURE & HIV MASTER

DNA/RNA, retro, CD4, window period

A+

Virology Base

HIV PRIORITY

Core: Viruses = obligate intracellular, filterable, no agar growth — cell/tissue culture or PCR. HIV = RNA retro, reverse transcriptase, CD4 killer.

Topic	Detail	Nursing cue
Shape	DNA (most stable) vs RNA (mutates); enveloped (fragile) vs naked (hardy)	Enveloped killed by soap/alcohol — hand hygiene works
HIV structure	gp120 + gp41 envelope, p24 core, RT enzyme	p24 = early antigen before antibodies
Transmission	Blood, sexual, vertical, NSI; NOT by hug/mosquito/sharing food	Universal precautions + PEP within 72 hrs (best 2 hrs)
Diagnosis	NACO: screening (ELISA/Rapid) → supplemental → confirmatory; PCR for infants <18 mo	Window 2-12 wks — counsel, repeat, no false reassurance
CD4 + ART	CD4 <350 start (test-treat now); viral load monitoring	TLD regimen; adherence >95%; no breastfeeding if safe replacement (as per current guideline box)

FATAL TRAP: Declaring HIV negative on one rapid test in window period. Repeat after window + PCR for infants (maternal antibodies persist).

NORCET HIT: p24 early, antibody late, PCR for babies.

Speed Check: Enzyme defining retrovirus?

Reverse transcriptase — RNA to DNA.

HEPATITIS VIRUSES A TO E TABLE

Route, carrier, vaccine in one matrix

A+

HEPATITIS MATRIX

VACCINE CLUE

Core: A+E = fecal-oral (no carrier, vaccine A); B+C+D = blood/sexual (carrier + chronic, vaccine B protects D too).

Virus	Route / Carrier	Vaccine / Marker
HAV	Fecal-oral, shellfish; NO carrier, self-limiting	Vaccine yes; IgM anti-HAV acute
HBV	Blood/sexual/vertical; CARRIER + chronic; Dane particle	Vaccine yes (0-1-6); HBsAg carrier, anti-HBs immune
HCV	Blood/NSI; chronic → cirrhosis; NO vaccine	Anti-HCV + RNA PCR; DAA cure
HDV	Defective — needs HBV helper	HBV vaccine protects D too
HEV	Fecal-oral; dangerous in PREGNANCY (fulminant)	No routine vaccine; safe water

FATAL TRAP: Forgetting HEV kills pregnant women. Highest fulminant failure in 3rd trimester — admit, monitor PT/encephalopathy.**NORCET HIT:** A+E gut, B+C blood. B vaccine blocks B+D.**Speed Check: Marker of HBV immunity after vaccine?**

Anti-HBs positive, HBsAg negative.

RABIES, POLIO, MEASLES & DIAGNOSIS

Negri bodies, AFP, Koplik + lab methods

A+

VIRUS CLINIC

PEP + SAMPLE

Core: Rabies = 100% fatal once symptoms, 100% preventable by wound wash + PEP. Polio = AFP surveillance. Measles = Koplik + Warthin-Finkeldey.

Virus	Clue / Test	Prevention / Care
Rabies (rhabdo)	Negri bodies in hippocampus; hydrophobia, aerophobia	Wash 15 min soap-water + povidone; Category III: RIG + ARV (0-3-7-14-28 Essen); no suturing first
Polio (entero)	AFP in <15 yrs; stool x2 in 14 days; OPV vs IPV	Pulse polio, cold chain, AFP reporting within 24 hrs
Measles / Rubella	Koplik spots, Warthin cells; CRS in pregnancy	MR vaccine; airborne; vitamin A in measles protocol
Lab diagnosis	PCR (gold), ELISA IgM, culture (cytopathic), Negri/Guarnieri bodies	Transport in VTM on ice; never formalin for PCR

FATAL TRAP: Suturing dog-bite wound immediately + skipping wash. Wash first, RIG local infiltration, delayed minimal suturing.

NORCET HIT: Wash-Watch-Vaccinate for every bite. Negri = Rabies.

Speed Check: FIRST aid for Category III dog bite?

Wash 15 min with soap + running water, then RIG + ARV.

FUNGI: SUPERFICIAL TO DEEP

KOH to capsule — site decides

A**MYCOSES MAP****SDA + KOH****Core:** Superficial = skin/hair/nail (dermatophytes); Subcutaneous = thorn prick; Systemic = lung; Opportunistic = low immunity.

Type	Organism / Clue	Test / Drug cue
Dermatophytes	Trichophyton, Microsporum, Epidermophyton; ringworm, athlete foot	KOH + SDA; griseofulvin/terbinafine; dry lesions
Subcutaneous	Sporothrix (rose gardener), Madura foot, chromoblastomycosis	Biopsy + LPCB; itraconazole
Systemic dimorphic	Histoplasma, Blastomyces, Coccidioides (25 mould, 37 yeast)	Dual-temp culture; amphotericin
Opportunistic	Candida, Aspergillus, Mucor, Crypto, Pneumocystis	CD4 low, steroids, ICU — see next page

FATAL TRAP: Applying steroids alone to ringworm. Tinea incognito spreads — antifungal first, no blind steroid cream.**NORCET HIT:** Ring + Scale = Dermatophyte. Rose thorn = Sporothrix.**Speed Check: Medium + temp for dermatophytes?**

SDA at 25°C — acidic, room temp for mould phase.

CANDIDA, ASPERGILLUS, MUCOR, CRYPTO

ICU opportunists every NORCET asks

A+**OPPORTUNIST CARD****GERM TUBE + GALACTO**

Core: Candida = germ tube; Aspergillus = acute-angle hyphae + galactomannan; Mucor = wide-angle + black necrotic eschar; Crypto = capsule.

Fungus	Clue / Test	Risk / Care
Candida albicans	Germ tube +, pseudohyphae, curd-white patches	Broad antibiotics, diabetes; fluconazole; nystatin for thrush
Aspergillus fumigatus	Acute 45° septate hyphae; air-crescent; galactomannan +	Neutropenia, construction dust; voriconazole; HEPA room
Mucor / Rhizopus	Broad aseptate 90° hyphae; black turbinate/eschar; reverse halo	DKA, steroids, post-viral; liposomal amphotericin + debridement
Cryptococcus neoformans	India ink halo; latex antigen; pigeon droppings	HIV low CD4 meningitis; fluconazole prophylaxis

FATAL TRAP: Missing black eschar in DKA nose as Mucor. Delay kills — urgent ENT, amphotericin, sugar control, no nasal steroids.

NORCET HIT: Germ tube = Candida, Acute = Aspergillus, Wide + Black = Mucor.

Speed Check: Test proving C. albicans in 2 hours?

Germ tube in serum at 37°C.

PARASITES: MALARIA TO AMOEBA

Protozoa first — vector + stool

A

PROTOZOA CARD

THICK + THIN SMEAR

Core: Protozoa = single cell (Plasmodium, Entamoeba, Giardia); diagnosed by blood smear / stool O&P; vectors decide control.

Parasite	Vector / Test	Disease / Care
Plasmodium (vivax/falciparum)	Female Anopheles; thick (detect) + thin (species); RDT HRP2/pLDH	Cyclic chill-fever-sweat; falciparum blackwater + cerebral; ACT + primaquine (G6PD check)
Entamoeba histolytica	Fecal-oral; stool with flask ulcer, anchovy-sauce liver abscess	Bloody mucoid stools; metronidazole + diloxanide; safe water
Giardia / Trichomonas	Fecal-oral / sexual; falling-leaf motility; strawberry cervix	Frothy diarrhoea / green discharge; metronidazole both partners
Leishmania / Trypanosoma	Sandfly / tsetse; LD bodies, Kala-azar splenomegaly	Sodium stibogluconate; vector control

FATAL TRAP: Giving primaquine without G6PD. Causes hemolysis — test G6PD first, especially radical cure vivax.**NORCET HIT:** Thick = find, Thin = type. Falciparum = deadly.**Speed Check: RDT antigen for falciparum?**

HRP2 — plus pLDH pan-malaria.

HELMINTHS, RODENTS & VECTORS

Worms + rat + mosquito triangle

A

HELMINTH + VECTOR

STOOL O&P

Core: Helminths = worms (round, tape, fluke). Rodents spread plague/lepto/rat-bite. Vector control = nurse health talk.

Worm / Vector	Clue	Care / Prevention
Ascaris / Hookworm / Enterobius	Bile-stained ova; ground itch + anemia; night itching	Albendazole + iron; footwear, handwash, nail trim
Taenia / Echinococcus	Pork/beef measly; hydatid cyst (water-lily)	Praziquantel; meat inspection; no raw salad in endemic
Filaria (Wuchereria)	Female Culex; night blood smear microfilaria	DEC + limb elevation; MDA programme
Rodents/vectors	Rat flea-plague; Leptospira-urine; Aedes-dengue; Anopheles-malaria	BMW + food cover, trap, ULV fogging, source reduction

FATAL TRAP: Day blood sample for filaria. Microfilaria peaks night — take night smear 10pm-2am.**NORCET HIT:** Night blood = Filaria. Night itch = Threadworm.**Speed Check: Drug + timing for filaria smear?**

DEC; night blood 10pm-2am for microfilaria.

LAB ID OVERVIEW FOR ALL PATHOGENS

Specimen → test → appearance in one shot

A+

ID MATRIX

SPECIMEN FIRST

Core: Pus = Gram + culture; sputum = ZN + CBNAAT; stool = O&P + hanging drop; blood = culture + smear + serology.

Specimen	First test	Confirms
Pus / wound	Gram + aerobic/anaerobic culture	Staph/Strep/Clostridium
Throat / nasal	Albert / culture on Loeffler + ASLO	Diphtheria / Strep carriage
Sputum	ZN + CBNAAT + LJ	TB vs bacterial pneumonia
Stool	O&P + hanging drop + TCBS/XLD	Amoeba/Giardia/Vibrio/Salmonella
Blood	Culture + thick/thin + Widal/ELISA	Typhi/malaria/HIV/hepatitis
CSF / skin	India ink + Gram/ZN + KOH	Crypto/meningococcus/fungus

FATAL TRAP: Sending stool in urine container or urine in stool pot.
Container decides rejection — sterile wide-mouth for urine, clean dry for stool.

NORCET HIT: Label bedside with antibiotics status or lab rejects.

Speed Check: Sputum test when ZN negative but TB suspected?

CBNAAT + LJ culture — never rule out on smear alone.

TRANSMISSION & PORTAL MAP

Source → portal → nursing break

A+

CHAIN BREAKER

ISOLATION LINK

Core: Know source + portal to pick isolation + PPE (links to Batch 4). Droplet vs airborne vs contact starts here.

Organism	Source / Portal	Break (preview)
Staph / Strep	Skin/nasopharynx, contact/droplet	Hand hygiene + wound barrier + contact
TB / Measles / Chickenpox	Airborne droplet nuclei	N95 + negative pressure (Batch 4)
Cholera / Typhi / Polio	Fecal-oral, water/food	Safe water + handwash + ORS
HIV / HBV / Malaria	Blood/sexual/vector	Safe needle + condoms + bed nets
Tetanus / Gas gangrene	Soil/manure via wound	Wound cleaning + TT + sterile delivery

FATAL TRAP: Putting TB patient on droplet mask only. Needs airborne N95 + negative pressure, not surgical mask.**NORCET HIT:** Portal decides PPE — learn now, apply in Batch 4.**Speed Check: Portal for tetanus spores?**

Wound/soil entry — never droplet.

BATCH 2 BLITZ (18 IN 3 MIN)

Cocci, bacilli, viruses, fungi, parasites

A+**SPEED REVISION****18 Q&A**

#	Q	A
1	Coagulase + clusters?	S. aureus
2	Catalase - chains?	Strep pyogenes
3	Optochin + lancet?	Pneumo
4	Oxidase + bean?	Neisseria
5	Bull neck membrane?	Diphtheria
6	Drumstick rigid?	Tetani
7	Swarming + staghorn?	Proteus
8	Widal week?	Week 2
9	Rice-water?	Cholera
10	Medusa-head?	Anthrax
11	Retro enzyme?	Reverse transcriptase
12	HEV danger?	Pregnancy fulminant
13	Negri bodies?	Rabies
14	Germ tube?	Candida albicans
15	Black eschar DKA?	Mucor
16	Thick vs thin?	Find vs type malaria
17	Night blood?	Filaria
18	Next Batch 3?	Immunity, Ig, vaccines

Next: Batch 3 = Section A Unit IV — Immunity tree, antigen-antibody, Ig, hypersensitivity, vaccines + cold chain.

IMMUNITY TREE IN ONE PAGE

Innate vs acquired, active vs passive

A+

TREE CARD

INC MICR 201

Core: Innate = birth, non-specific. Acquired = after exposure, specific, memory. Active = self-made (slow, lasting). Passive = ready-made (fast, short).

Type	Examples	Nursing cue
Innate: barriers	Skin, cilia, gastric acid, lysozyme, commensals	Intact skin + mucosa = first defence
Innate: cells	Neutrophils, macrophages, NK cells, complement, interferon	Pus = neutrophils at work
Active natural	After infection (measles lifelong)	History of disease = immunity
Active artificial	Vaccines (BCG, DPT, OPV)	Slow 2-4 wks, booster needed
Passive natural	IgG placenta, IgA breast milk	Protects 3-6 months only
Passive artificial	TIG, RIG, antivenom, immunoglobulin	Immediate, post-exposure, short

FATAL TRAP: Vaccines are ACTIVE not passive. Immunoglobulin/antitoxin is PASSIVE. MCQ swaps them.

NORCET HIT: Self-made = Active lasting. Ready-made = Passive passing.

Speed Check: Placental antibody class?

IgG only crosses placenta — protects newborn first months.

NON-SPECIFIC KILLERS: PHAGOCYTOSIS TO IFN

Complement, inflammation, fever

A+

MECHANISM

Core: Phagocytosis (neutrophil/macrophage) + complement lysis + interferon antiviral + inflammation tetrad.

- **Phagocytosis steps:** chemotaxis → attachment (opsonin) → engulf → phagolysosome kill → exocytosis.
- **Complement (C1-C9):** opsonization, chemotaxis, MAC lysis; activated by classical (antibody) or alternate.
- **Inflammation:** rubor, tumor, calor, dolor + functio laesa; CRP rises.
- **Interferon-alpha/beta:** virus-infected cell warns neighbours — antiviral state, not virus-specific.
- **Fever:** IL-1/pyrogen resets hypothalamus — enhances immunity, inhibits microbes.

Marker	Means	Bedside
Neutrophilia	Acute bacterial	Pus, high TLC
Lymphocytosis	Viral/chronic	TB, whooping cough
Eosinophilia	Parasite/allergy	Worms, asthma

FATAL TRAP: Giving antipyretic to mask fever before blood culture. Fever guides sepsis workup — culture first.

NORCET HIT: Pus = neutrophil graveyard. Granuloma = macrophage wall.

Speed Check: Cells of pus?

Dead neutrophils + bacteria + debris.

ANTIGEN, ANTIBODY & SEROLOGY BASE

Agglutination to ELISA principle

A

REACTION TYPES

Core: Antigen = foreign trigger. Antibody = Y-shaped Ig made by plasma cells. Reaction visible as clump/lysis/colour.

Reaction	Example test	Use
Agglutination	Widal, blood grouping, Coombs	Particulate clumping
Precipitation	ELEK (diphtheria), VDRL flocculation	Soluble lines
Complement fixation	Historical TB/viral	Lysis = positive
ELISA / Rapid	HIV, HBsAg, dengue NS1/IgM	Colour = antigen/antibody present
Neutralization	ASLO, antitoxin	Toxin blocked

- **Prozone trap:** excess antibody gives false negative — dilute serum.
- **Paired sera:** 4-fold rise 2 weeks apart proves recent infection.

FATAL TRAP: Single Widal 1:80 = typhoid. Needs rising titre + clinical + culture. Treat patient, not paper.

NORCET HIT: Clump = agglutination (Widal). Colour = ELISA.

Speed Check: Test for blood group compatibility?

Agglutination — clumping means incompatibility.

IGG & IGM: THE CORE TWO

Placenta vs first responder

A+

Ig MATRIX

Core: Y = 2 heavy + 2 light chains. IgG most abundant, only placental. IgM largest (pentamer), first in infection.

Ig	Structure / Location	Classic clue
IgG (75%)	Monomer, serum, crosses placenta, opsonizes, complements	Past infection / vaccine immunity; Rh anti-D is IgG
IgM (10%)	Pentamer, intravascular, first antibody, B-cell receptor	Acute infection (TORCH IgM = recent); blood group ABO natural
Serology rule	IgM = now, IgG = past/protected	Dengue NS1 early, IgM day 5, IgG later

FATAL TRAP: IgM in newborn = fetal infection (cannot cross placenta).
TORCH IgM positive baby = congenital infection, not maternal transfer.

NORCET HIT: M = Mega + Moment (first). G = Goes across placenta +
Greatest amount.

Speed Check: Antibody proving recent dengue?

IgM positive — first responder of acute infection.

IGA, IGE, IGD IN 5 MINUTES

Mucosa, allergy, B-cell

A+

Ig FINISHER

Ig	Location / Function	Disease link
IgA (15%)	Secretory: breast milk, saliva, tears, gut; dimer + J chain + secretory piece	Protects mucosa; deficient = recurrent sinopulmonary + diarrhoea
IgE (<1%)	Mast cell/basophil bound; helminth + allergy mediator	Asthma, anaphylaxis, hay fever; high in worms
IgD	B-cell surface with IgM; starts response	Marker of naive B cells — rarely tested

- **Colostrum = IgA:** gut protection, not blood. Breastfeed within 1 hour.
- **Allergy workup:** high IgE + eosinophils = atopy/worms.

FATAL TRAP: IVIG is IgG pool — anaphylaxis risk in IgA-deficient patient (anti-IgA antibodies). Screen before transfusion.

NORCET HIT: A = Airway/milk surface. E = Eosinophil + Emergency allergy.

Speed Check: Antibody in colostrum?

Secretory IgA — mucosal protection of newborn gut.

TYPE I: ANAPHYLAXIS EMERGENCY

IgE + mast cell + adrenaline

A+

EMERGENCY CARD

Core: Prior sensitization → re-exposure → IgE-mast degranulation (histamine) → vasodilation + bronchospasm in minutes.

- **Triggers:** penicillin, insect sting, latex, blood, foods (peanut), contrast.
- **Signs:** itch + urticaria + angioedema + wheeze + hypotension + tachycardia.
- **FIRST drug:** IM Adrenaline 0.5 mg (0.5 mL 1:1000) mid-outer thigh, repeat 5 min; high-flow O₂, legs up, IV fluids, nebulized salbutamol, chlorpheniramine + hydrocortisone later.
- **Nursing:** stop trigger infusion, keep airway kit, never IV bolus adrenaline peripherally outside protocol.

FATAL TRAP: Giving antihistamine alone for anaphylaxis. Adrenaline first — antihistamine never saves airway/hypotension.

NORCET HIT: Exposure + minutes + wheeze + low BP = Type I. Thigh IM adrenaline.

Speed Check: Dose/route of first adrenaline?

0.5 mg IM antero-lateral thigh (1:1000), repeat q5min.

TYPE II & III: CYTOTOXIC + IMMUNE COMPLEX

Blood reactions + serum sickness

A+

COMPARE

Type	Mechanism	Examples
II Cytotoxic (IgG/IgM)	Antibody vs cell surface → lysis/phagocytosis	Transfusion reaction, HDN, myasthenia, Goodpasture, ITP
III Immune complex	Ag-Ab deposits in vessels/kidney/joints → complement	SLE, PSGN, serum sickness, Arthus, farmer lung

- **Transfusion reaction cue:** fever + chills + flank pain + red urine (hemolysis) → STOP transfusion, saline, notify, return bag.
- **PSGN cue:** cola urine + edema + HTN 1-3 wks post strep throat.

FATAL TRAP: Continuing transfusion despite reaction. Stop first, keep IV with saline, save bag + sample.

NORCET HIT: II = vs Cells. III = Complex deposits in kidney/joint.

Speed Check: FIRST action in transfusion reaction?

STOP transfusion immediately, run normal saline, call doctor.

TYPE IV: DELAYED T-CELL (48-72 HRS)

Mantoux, contact dermatitis, granuloma

A+

DELAYED CARD

Core: No antibody. Sensitized T-cells + macrophages → induration in 48-72 hrs. No wheal within minutes.

Test / Disease	Reading	Nursing cue
Mantoux (TST)	0.1 mL ID PPD, read 48-72 hrs induration (not redness)	≥10 mm positive (≥5 mm if HIV/contact); BCG cross-reacts
Contact dermatitis	Nickel, latex, neomycin — eczema 2-3 days	Patch test; remove trigger, topical steroid
Granuloma (TB/leprosy)	Caseating vs non-caseating	Type IV walling off bacilli
Transplant rejection (cellular)	Days-weeks	Tacrolimus/cyclosporine suppress T-cells

FATAL TRAP: Reading Mantoux at 20 minutes like allergy. Must wait 48-72 hrs and measure induration, not erythema.

NORCET HIT: Minutes = I, Hours-cytotoxic = II, Complex = III, Days = IV.

Speed Check: When to read Mantoux?

48-72 hours, transverse induration in mm.

ALL 4 TYPES: ONE REVISION TABLE

Mediator, time, example

A+

MASTER TABLE

Type	Mediator / Time	Classic examples
I Anaphylactic	IgE, minutes	Anaphylaxis, asthma, hay fever, food allergy
II Cytotoxic	IgG/IgM, hours	Transfusion, HDN, Goodpasture
III Complex	Ag-Ab + complement, hours-days	SLE, PSGN, serum sickness
IV Delayed	T-cells, 48-72 hrs	Mantoux, contact dermatitis, TB granuloma

Mnemonic ACID: Anaphylactic-Cytotoxic-ImmuneComplex-Delayed. Time: Minutes-Hours-Days for I-II/III-IV.

Speed Check: Serum sickness type?

Type III immune-complex — fever + rash + joint pain 7-10 days after antiserum.

LIVE VS KILLED VACCINES

Rule + handling

A+

VACCINE TREE

Class	Examples	Rules
Live attenuated	BCG, OPV, MMR, rotavirus, yellow fever, intranasal flu	Strong, single dose often; avoid in pregnancy + severe immunodeficiency; cold chain strict
Killed/inactivated	IPV, rabies, HAV, whole-cell pertussis, Covaxin-type	Safe in pregnancy/HIV; needs boosters
Subunit/toxoid/conjugate	DPT (toxoid), Hib, pneumo, HepB, HPV	Very safe; multiple doses
mRNA (newer)	COVID mRNA types	No live virus; cold chain ultra

FATAL TRAP: Live vaccine in pregnancy/advanced HIV. Killed first — live contraindicated (except as per current programme exception boxes).

NORCET HIT: Live = BCG-OPV-MMR. Killed = R-I-P (Rabies, IPV, Pertussis-wc).

Speed Check: Which polio vaccine sheds + needs cold chain most?

OPV live — keep 2–8°C, VVM watch.

BCG, OPV, DPT, MMR: SITE + ROUTE

Bedside administration card

A+**SITE-ROUTE**

Vaccine	Site / Route / Time	After-care
BCG (birth)	Left upper arm ID 0.05 mL (0.1 after 1 mo); wheal must form	Scar at 6-12 wks normal; no antiseptic, no massage; lymphadenitis report
OPV (birth, 6-10-14 wks + pulse)	Oral 2 drops; VVM + cold chain	No hot milk 30 min; AFP watch
Pentavalent/DPT (6-10-14 wks)	Antero-lateral thigh IM; no gluteal in infants	Fever/ swelling counselling; next dose date card
MMR/MR (9 + 15 mo)	Right upper arm SC	Rash/fever 7-10 days normal; cold + paracetamol guidance

FATAL TRAP: Cleaning BCG site with spirit + rubbing. Kills live bacilli, no scar, no immunity. Dry sterile technique.

NORCET HIT: Left arm BCG-ID, Thigh DPT-IM, Right arm MMR-SC, Mouth OPV.

Speed Check: Normal BCG evolution?

Papule → ulcer → scar 6-12 weeks. No antibiotics needed.

COLD CHAIN & VVM MASTER

2-8°C, ILR, VVM stages

A+**COLD CHAIN**

- **Range:** +2 to +8°C (ILR); freezer -15 to -25°C for OPV stock. No freezing of DPT/HepB/TT (shake test).
- **VVM:** Stage 1 (light) usable → Stage 2 usable → Stage 3-4 discard. Inner square darker than outer = discard.
- **Placement:** BCG/measles top, OPV middle, DPT/HepB lower basket away from cooling plates; thermometer twice daily.
- **Open vial policy:** check current programme time-box (multi-dose vs BCG/measles discard rule) — write date-time on label.
- **Power cut:** keep door closed, ice packs conditioned, transfer log, never use potency-doubt vial without officer decision.

FATAL TRAP: Using frozen DPT/HepB (flocculated). Shake test granular = discard. Freezing destroys potency irreversibly.

NORCET HIT: Freeze kills DPT, heat kills OPV/BCG. VVM watches heat.

Speed Check: VVM discard point?

Inner square matches/darker than outer ring — discard.

AEFI WATCH & SCHEDULE BOX

What to record + when to refer

A+**SAFETY BOX**

- **Minor:** pain, low fever, rash — paracetamol, cold compress, reassure, document.
- **Severe/AEFI:** anaphylaxis, seizure, HHE, abscess — adrenaline kit, referral, AEFI form within 24 hrs, keep vial.
- **4-key recording:** date-time, vaccine + batch, site-route, next due + mother signature.
- **Schedule box (updateable):** Birth BCG+OPV+HepB → 6/10/14 wks Penta+OPV → 9 mo MR+VitA → 15 mo MMR → 16-24 mo DPT booster → 5-6 yrs DPT → 10/16 yrs TT/Td. Verify against current UIP card before exam.

FATAL TRAP: Missing AEFI documentation. Every severe event needs batch no. + VVM + time + action — medico-legal must.

NORCET HIT: Card + Batch + Next date = safe vaccinator.

Speed Check: First record after any vaccine?

Batch no., site-route, time, next due on card + register.

DENGUE, WIDAL, HIV TESTING TRAPS

Day-wise markers

A

DAY-WISE

Day	Test	Meaning
Dengue 1-5	NS1 antigen	Early positive before antibodies
Dengue 5+	IgM then IgG	Recent vs past; tourniquet + warning signs
Typhoid wk1/wk2	Blood culture / Widal	Culture early, antibodies late
HIV window	p24 / RNA then ELISA	Repeat after window; infant PCR

Speed Check: Earliest dengue marker?

NS1 antigen day 1-5.

COMPLEMENT PATHWAYS & INTERFERONS

Classical vs alternate

A**PATHWAY CARD**

- **Classical:** Ag-Ab → C1 → MAC lysis. **Alternate:** endotoxin surface directly → C3.
- **Functions:** lysis, opsonin (C3b), chemotaxis (C5a), anaphylatoxin.
- **Interferons:** alpha (leukocyte), beta (fibroblast), gamma (T-cell) — antiviral + macrophage boost.
- **Low C3/C4 cue:** active SLE / PSGN consuming complements.

Speed Check: Opsonin fragment?

C3b — coats microbe for phagocytosis.

PRIMARY VS SECONDARY FAILURE

SCID to AIDS bedside

A

FAILURE MAP

Type	Examples	Nursing cue
Primary (genetic)	SCID, Bruton (no B), DiGeorge (no T)	Recurrent infections infancy; live vaccines danger; isolation
Secondary	HIV, steroids, chemo, malnutrition, diabetes	Opportunistic screen; hand + mask + reverse barrier as ordered

Speed Check: Bruton defect?

X-linked agammaglobulinemia — no B cells, no antibodies.

GRAFT REJECTION & IMMUNOSUPPRESSION

Hyperacute to chronic

A**LINK CARD**

- **Hyperacute (minutes):** preformed ABO antibodies — check group first.
- **Acute (days-weeks):** T-cell — tacrolimus/cyclosporine + steroids.
- **Chronic (months-years):** vascular fibrosis — BP/lipid control.
- **Nursing:** strict intake-output, infection watch, no live vaccines, adherence counselling.

Speed Check: Drug blocking T-cell IL-2?

Cyclosporine/tacrolimus (calcineurin inhibitors).

BATCH 3 BLITZ (18 IN 3 MIN)

Immunity to vaccines

A+**18 Q&A**

#	Q	A
1	Placental Ig?	IgG
2	First Ig?	IgM
3	Mucosal Ig?	IgA
4	Allergy Ig?	IgE
5	Type I drug?	IM adrenaline thigh
6	Transfusion stop?	Type II — stop + saline
7	PSGN type?	Type III
8	Mantoux read?	48-72h induration
9	Live vaccines?	BCG OPV MMR
10	BCG site?	Left arm ID
11	Cold range?	2-8°C
12	VVM discard?	Inner darker than outer
13	NS1 when?	Dengue day 1-5
14	Opsonin?	C3b
15	Bruton?	No B cells
16	Rejection drug?	Tacrolimus
17	Passive artificial?	TIG/RIG/antivenom
18	Next?	HAI + isolation/PPE

HAI CONCEPT & INFECTION CONTROL TEAM

48-hr rule, ICT, surveillance

A+

HAI BASE

Core: HAI = infection after 48 hrs admission, not incubating on arrival. ICT = microbiologist + nurse (ICN) + doctor + housekeeping.

HAI	Cutoff / Site	Device link
CAUTI	UTI after 48h with catheter	Catheter days denominator
CLABSI	Bloodstream with central line 48h+	Line days
SSI	30 days (1 yr implant) post-op	Shave, prophylaxis timing
VAP/VAE	Ventilator 48h+ pneumonia/event	Vent days, head elevation

- **Committee:** HICC meets monthly; ICN daily rounds, bundle audit, hand-hygiene compliance.
- **Rate formula:** (infections/device-days) x 1000. Zero target.

FATAL TRAP: Calling admission-day UTI as CAUTI. Needs 48h + device + no prior incubation.

NORCET HIT: 48 hrs + device-days x1000 = HAI math.

Speed Check: HAI time cutoff?

After 48 hours of admission.

CAUTI PREVENTION BUNDLE

Insert only if needed, remove fast

A+**BUNDLE CARD**

- **Avoid:** no routine incontinence catheter; condom catheter, bladder scan first.
- **Aseptic insertion:** sterile tray, peri-clean, small bore, balloon per size, secure to thigh.
- **Closed drainage:** never disconnect; bag below bladder, off floor; empty with clean gloves, no touch spigot.
- **Daily review:** necessity + nurse-driven removal protocol; document indication.
- **Hygiene:** peri-care soap-water daily; no routine antiseptic irrigation.

Do	Do NOT
Secure, unobstructed flow, daily need check	Irrigate prophylactically, open circuit, bag on bed

FATAL TRAP: Routine bladder wash for prevention. Increases CAUTI — prohibited.

NORCET HIT: Closed + Low + Daily review = CAUTI zero.

Speed Check: Bag position?

Below bladder, off floor, no kinks.

CLABSI: LINE INSERTION TO CARE

Maximal barrier + scrub the hub

A+**LINE CARD**

- **Insertion:** hand hygiene, maximal barrier (cap-mask-gown-gloves-drape), 2% chlorhexidine-alcohol, avoid femoral, ultrasound.
- **Maintenance:** daily need, scrub hub 15 sec alcohol, sterile dressing change (gauze 2 days, transparent 7 days or soiled/loose), date-label.
- **Watch:** fever + line + no other source = suspect; paired cultures (line + peripheral) before antibiotics.
- **Remove:** prompt removal when not needed; no routine replacement.

FATAL TRAP: Changing line over guidewire for infection. Remove + new site, never exchange infected line.

NORCET HIT: Scrub hub 15 sec every access.

Speed Check: Skin prep for central line?

2% chlorhexidine in 70% alcohol, dry fully.

SSI PREVENTION: BEFORE TO AFTER CUT

Bath, clip, prophylaxis timing

A+**SSI CARD**

- **Pre-op:** chlorhexidine bath night + morning, clip (no razor), control sugars, stop smoking, no prolonged pre-op stay.
- **Antibiotic:** within 60 min incision (120 for vancomycin), re-dose long cases, stop within 24h (48 cardiac).
- **Intra-op:** normothermia, glycemia, O₂, traffic control, sterile technique.
- **Post-op:** sterile dressing 24-48h, wound class reporting, surveillance 30d/1yr.

FATAL TRAP: Shaving night before with razor. Micro-cuts seed SSI — clip just before, no razors.

NORCET HIT: Clip not shave, 60-min antibiotic, 24-h stop.

Speed Check: SSI window with implant?

1 year surveillance.

VAP/VAE: VENTILATOR BUNDLE

Head up, sedation break, oral care

A+**VENT BUNDLE**

- **Head 30-45°**, sedation vacation + weaning assess daily, DVT + ulcer prophylaxis.
- **Oral care:** chlorhexidine + subglottic suction, cuff 20-30 cmH₂O, closed suction.
- **Prevent:** hand hygiene, no routine circuit change unless soiled, HME change protocol.
- **Watch:** new fever + purulent + infiltrate + rising O₂ need after 48h vent = VAE.

FATAL TRAP: Flat supine ventilated patient. Aspiration + VAP — keep 30-45° unless contraindicated.

NORCET HIT: HEAD-UP + BREAK + BRUSH = VAP down.

Speed Check: Cuff pressure?

20-30 cmH₂O (20-25 mmHg) — seal without ischemia.

SURVEILLANCE & DEVICE-DAY MATH

Numerator, denominator, action

A+

RATE CARD

Rate	Formula	Target
CAUTI/1000 catheter-days	Infections/days x1000	Zero, review insertion
CLABSI/1000 line-days	Same	Zero, hub audit
VAP/1000 vent-days	Same	Zero, bundle audit
SSI %	SSI/operations x100	Benchmark, feedback to surgeon

- **Active vs passive:** ICN ward rounds + lab alerts vs reports only.
- **Outbreak:** 2+ linked cases → cohort, reinforce, inform HICC.

Speed Check: Denominator for CLABSI?

Central-line days.

STANDARD PRECAUTIONS FOR ALL

Assume every fluid infectious

A+**UNIVERSAL BASE**

- **Applies to:** blood, all body fluids/secretions/excretions (except sweat), non-intact skin, mucous membranes — every patient, every time.
- **Includes:** hand hygiene, PPE per risk, safe sharps, cough etiquette, clean equipment, linen + waste + laundry routine.
- **Risk assessment:** choose mask/gown/eye protection by splash/spray anticipation, not diagnosis.

FATAL TRAP: Gloves replace hand hygiene. Hands before + after gloves, always.

NORCET HIT: Standard = Everyone + Every fluid (minus sweat).

Speed Check: Only fluid excluded?

Sweat (unless visibly bloody).

CONTACT PRECAUTIONS: MRSA, VRE, C. DIFF

Gown + gloves + private/cohort

A+**CONTACT CARD**

- **Diseases:** MRSA, VRE, MDR, scabies, C.difficile, norovirus, wound infections.
- **Room:** single/cohort; dedicated equipment (stethoscope, BP cuff).
- **PPE:** gown + gloves on entry, discard before exit, handwash (soap-water for C.diff/noro — alcohol fails spores).
- **Transport:** cover wounds, clean wheelchair, inform receiving unit.

FATAL TRAP: Alcohol rub for C.diff. Spores resist — soap-water wash mandatory.

NORCET HIT: Contact = Gown + Gloves + Gear dedicated.

Speed Check: Hand method for C.diff?

Soap-water wash, not alcohol rub.

DROPLET: FLU, MENINGITIS, MUMPS

3-feet surgical mask rule

A+**DROPLET CARD**

- **Diseases:** influenza, meningococcal, pertussis, mumps, rubella, diphtheria, COVID (droplet + contact context).
- **Distance:** large particles >5 micron fall within 3 feet/1 m — surgical mask within 3 feet, private/cohort, door may stay open.
- **Patient:** mask during transport; cough etiquette; spatial separation.

FATAL TRAP: N95 for droplet only. Surgical mask suffices within 3 feet — save N95 for airborne.

NORCET HIT: Droplet = 3 feet + Surgical mask.

Speed Check: Door for droplet room?

May remain open; negative pressure NOT required.

AIRBORNE: TB, MEASLES, CHICKENPOX

N95 + negative pressure

A+

AIRBORNE CARD

- **Diseases:** TB, measles, chickenpox/disseminated zoster — nuclei <5 micron float hours.
- **Room:** negative pressure AIIR, 6-12 air changes, door CLOSED, exhaust outside/HEPA.
- **Mask:** fit-tested N95 for staff; patient wears surgical mask if transport.
- **Susceptibles:** immune staff only for measles/chickenpox; non-immune excluded.

FATAL TRAP: Surgical mask for TB care. Needs N95 + fit seal — surgical leaks.

NORCET HIT: Airborne = N95 + Negative + Closed door. Mnemonic M-T-C (Measles-TB-Chickenpox).

Speed Check: Air changes for AIIR?

6-12 per hour, negative pressure.

DONNING & DOFFING ORDER

Outside contaminated, inside clean

A+**SEQUENCE MUST****Don (enter)**

1 Gown → 2 Mask/N95 → 3 Goggles/face shield → 4 Gloves over cuffs

Doff (exit, before room door)

1 Gloves → 2 Goggles → 3 Gown → 4 Mask (outside room) + hand hygiene each step gap

- **Fit check:** N95 seal inhale/exhale, no beard gap; hand hygiene before don + after doff.
- **Inspect:** tears, expiry, ties; buddy check in high-risk.

FATAL TRAP: Removing mask first inside room. Most contaminated last — mask off outside after hand hygiene.

NORCET HIT: Don: Gown-Mask-Goggles-Gloves. Doff: Gloves-Goggles-Gown-Mask.

Speed Check: Gloves over or under cuffs?

Over cuffs — no wrist exposure.

WHICH PPE WHEN? RISK TABLE

Splash, aerosol, contact risk

A+**SELECTOR**

Situation	PPE
Blood draw, IV	Gloves + hand hygiene
Suction, childbirth splash	Gloves + gown + mask + eye shield
TB sputum induction	N95 + gown + gloves, AIIR
MRSA dressing	Gown + gloves, dedicated kit
C.diff diarrhoea	Gown + gloves + soap wash

Speed Check: Eye protection when?

Anticipated splash/spray to face.

ROOM SPECS & COHORTING

Private, cohort, neutral

A+

ROOM CARD

- **Single preferred:** cohort same organism only (never MRSA + C.diff together).
- **Distance:** 3 feet beds in cohort droplet; curtains not wall.
- **Equipment:** dedicated, clean between; minimal stock inside.
- **Signage:** precaution card on door + PPE station outside.

Speed Check: Cohort rule?

Same pathogen, same precautions only.

SCENARIO DRILLS: PICK PRECAUTION

Exam vignettes solved

A+

SCENARIO GRID

Vignette	Answer
Purpuric meningitis, droplet nuclei?	Droplet + chemoprophylaxis contacts
Cavitary TB coughing	Airborne AHR + N95
Watery C.diff post-clinda	Contact + soap wash + sporicidal clean
Measles rash + cough	Airborne, immune staff only
Influenza ward surge	Droplet + cohort + mask on patient transport

Speed Check: Meningococcal contacts need?

Chemoprophylaxis (as per current protocol) + droplet 24h.

TRANSPORT & VISITOR RULES

Mask patient, inform unit

A+

MOVEMENT CARD

- **Limit movement;** mask patient (surgical), cover wounds, clean trolley before/after.
- **Inform:** receiving X-ray/OT + lift staff; hand PPE to receiver.
- **Visitors:** teach hand + mask + gown, no sitting on bed, no symptomatic visitors, children restricted.
- **Clean:** disinfect high-touch after transfer.

Speed Check: Who wears N95 during TB transport?

Staff N95; patient surgical mask.

BUNDLE AUDIT: NURSE CHECKLIST

Daily tick + feedback

A**AUDIT CUE**

- **Ask daily:** device still needed? insertion date? dressing intact? bag below? head up? oral done?
- **Document:** indication, size, attempts, asepsis, review, removal plan.
- **Feedback:** run charts to ward; celebrate zero-days.

Speed Check: Best CLABSI reducer?

Daily line-necessity review + prompt removal.

LINK TO NSI & LINEN PREVIEW

Sharps + laundry safe

A+

LINK CARD

- **Sharps:** no recapping, puncture-proof container 3/4 full rule (detail Batch 6/9).
- **Linen:** bag at bedside, no shaking/hugging, colour-coded (detail Batch 6).
- **Spill:** gloves + cover + hypochlorite (detail Batch 5).

Speed Check: Shake linen?

Never — aerosols pathogens.

BATCH 4 BLITZ (18 IN 3 MIN)

HAI + precautions

A+**18 Q&A**

#	Q	A
1	HAI cutoff?	48h
2	CAUTI bag?	Below bladder
3	Scrub hub?	15 sec
4	SSI shave?	Clip, no razor
5	VAP head?	30-45 deg
6	CLABSI prep?	2% CHG-alcohol
7	Standard for?	All + all fluids minus sweat
8	C.diff hands?	Soap-water
9	Droplet distance?	3 feet + surgical
10	Airborne trio?	Measles TB Chickenpox
11	N95 when?	Airborne only
12	Don order?	Gown-Mask-Goggles-Gloves
13	Doff first?	Gloves
14	Cohort rule?	Same bug only
15	TB transport?	Staff N95, pt surgical
16	AIIR air?	6-12, negative
17	Cuff?	20-30 cmH2O
18	Next?	Hand hygiene + sterilization

5 MOMENTS & 3 TYPES OF HAND HYGIENE

WHO moments, when rub vs wash

A+

5 MOMENTS

Core: Clean hands save lives. 5 moments decide when; method (rub/wash/scrub) decides how.

Moment	Example
1 Before patient contact	Enter ward, shake hand
2 Before aseptic task	IV, catheter, dressing, suction
3 After body-fluid risk	Blood, urine, secretions, samples
4 After patient contact	Leave bedside
5 After surroundings	Bed rail, monitor, pump

Type	When	Time
Alcohol rub	Visibly clean hands, all 5 moments except soiled/C.diff	20-30 sec till dry
Soap wash	Visibly dirty, C.diff/noro, after toilet, before food	40-60 sec
Surgical scrub	OT: 2-5 min chlorhexidine/povidone + sterile towel	2-5 min

Speed Check: Rub vs wash time?

Rub 20-30 sec, Wash 40-60 sec.

6 STEPS TECHNIQUE MASTER

Palm to wrist, no miss

A+**STEPS CARD**

- 1 Palm to palm. 2 Back of hands. 3 Palm-palm fingers interlaced. 4 Backs fingers to palms. 5 Rotational thumbs. 6 Rotational fingertips + wrists.
- **Must:** remove watch/rings, short nails, no artificial nails for high-risk; dry fully (wet = recontamination).
- **Surgical extra:** keep hands above elbows, sterile towel one hand each side, gown + closed gloving.

FATAL TRAP: Missing thumbs + fingertips. Most missed areas = thumbs, fingertips, wrists — where germs hide.

NORCET HIT: Palm-Back-Lace-Bend-Thumb-Tip.

Speed Check: Hands position in surgical scrub?

Above elbows, dripping to elbows.

ALCOHOL RUB RULES & ERRORS

Volume, dry time, jewellery

A+**ERROR TRAPS**

- **Dose:** 3-5 mL palmful, cover all surfaces, rub till DRY (no waving/fan).
- **Do NOT use rub:** visibly soiled, C.diff/norovirus outbreak, after toilet, chemicals.
- **Errors:** short 5-sec rub, gloves instead of hygiene, long nails/polish, missed audit.
- **Promotion:** point-of-care bottles, audit + feedback, role modelling.

FATAL TRAP: Rub on C.diff diarrhoea hands. Spores survive alcohol — soap-water wash only.

NORCET HIT: Clean-looking = Rub. Dirty/C.diff = Wash.

Speed Check: Artificial nails in ICU?

Prohibited — harbour Gram-neg + fungi.

SPAULDING: CRITICAL-SEMI-NONCRITICAL

Item decides method

A+**SPAULDING MASTER**

Class	Item	Method
Critical (sterile tissue/vascular)	Scalpel, implant, needle, cardiac catheter	Sterilize: autoclave/ETO/plasma
Semi-critical (mucosa/non-intact skin)	Endoscope, laryngoscope, diaphragm, thermometer	High-level disinfection (glutaraldehyde/OPA)
Non-critical (intact skin/surround)	BP cuff, bed, linen, stethoscope	Low/intermediate disinfection, cleaning

FATAL TRAP: High-level disinfected scope for sterile cavity. Semi-critical method never suffices for critical use.

NORCET HIT: Critical = Sterilize. Mucosa = High-level. Skin = Low-level.

Speed Check: Endoscope class?

Semi-critical — high-level disinfection.

AUTOCLAVE: STEAM UNDER PRESSURE

121°C 15 psi 15 min

A+

GOLD STANDARD

- **Principle:** moist heat coagulates protein; pressure raises boiling point.
- **Standard:** 121°C, 15 psi, 15-20 min (wrapped); 134°C flash 3-5 min porous.
- **Do NOT autoclave:** heat-labile (plastics), oils/powders (dry heat), sharps with caps sealed tight.
- **Packing:** loose, indicator inside + outside, date, no overfill, dry cycle.

Indicator	Example	Reads
Chemical	Bowie-Dick, tape, strips	Each cycle exposed
Biological	Geobacillus stearothermophilus spores	Weekly, true kill proof

FATAL TRAP: Opening drum before drying. Wet packs wick germs — cool, dry, store closed.

NORCET HIT: 121-15-15 = Autoclave mantra. Spore = Geobacillus.

Speed Check: Biological indicator organism?

Geobacillus stearothermophilus spores.

DRY HEAT, ETO, PLASMA, RADIATION

What when autoclave fails

A+

METHOD SELECTOR

Method	For / Condition	Indicator
Dry heat (hot air oven)	Oils, powders, glass: 160°C 1h / 170°C 30min	Bacillus atrophaeus spores
ETO gas	Heat-labile plastics, catheters; aerate 8-12h (toxic)	Bacillus atrophaeus
Plasma (H2O2)	Endoscopes, electronics; fast, no aeration	Chemical + biological
Radiation/ filtration	Disposable syringes (gamma); air (HEPA); liquids (membrane)	Dosimeter / integrity test

FATAL TRAP: Using ETO item immediately. Residual gas burns tissue — aerate fully.

NORCET HIT: Oil = Dry heat. Plastic = ETO/Plasma.

Speed Check: Dry-heat spore?

Bacillus atrophaeus (subtilis).

HYPOCHLORITE, GLUTARALDEHYDE, PHENOLS

Concentration + contact time

A+

CHEMICAL TABLE

Agent	Use / Strength	Caution
Sodium hypochlorite	Blood spill 1% (10,000 ppm); floor 0.5%; fresh daily	Corrosive, inactivated by organic matter — clean first
Glutaraldehyde 2% / OPA	Endoscope high-level 20-45 min	Rinse sterile water; fumes — closed lid
Phenol/Lysol, chlorhexidine-alcohol	Floors / skin prep	Never on neonate skin large area; dry fully
70% alcohol, H2O2	Trolley, thermometer, hub	Flammable; contact time matters

FATAL TRAP: Pouring hypochlorite on blood without cleaning. Organic load neutralizes — remove spill bulk first, then disinfect.

NORCET HIT: Blood = 1% hypo fresh. Scope = Glut 20 min+.

Speed Check: Blood spill strength?

1% hypochlorite (10,000 ppm), 10-20 min contact.

SPILL MANAGEMENT & HOUSEKEEPING

Cover, clean, disinfect

A+**SPILL STEPS**

- **Small spill:** gloves + gown + mask-eye → cover with absorbent → 1% hypo 10-20 min → scoop with cardboard → bag yellow → mop + document.
- **Large/mercury:** cordon, call housekeeping, ventilate; never vacuum mercury.
- **Routine:** high-touch (rails, knobs, switches) thrice daily; terminal clean on discharge; colour-coded mops.

Speed Check: First cover for blood spill?

Absorbent + 1% hypochlorite, wait, then collect.

CSSD: DIRTY TO STERILE FLOW

One-way, no backtrack

A

FLOW MAP

- **Zones:** receipt/wash → pack → sterilize → sterile store → issue. One-way forward, negative to positive pressure.
- **Nurse cue:** check pack integrity, indicator, expiry, dryness before opening; FIFO.

Speed Check: Opened sterile pack rule?

Use immediately or reprocess — never re-store open.

SPECIMEN RULES: BEFORE ANTIBIOTICS

Right time, container, label

A+**PRE-ANALYTICAL**

- **Before antibiotics:** active site; sterile container; bedside label (name/ID/site/time/Ab status); transport <2h.
- **Reject:** unlabeled, leaking, dry swab for anaerobe, formalin for culture, 24h old urine at room temp.

Speed Check: Most common rejection cause?

Unlabeled / mislabeled container.

BLOOD, URINE, SPUTUM COLLECTION

Sets, midstream, deep cough

A+**TOP 3 SPECIMENS**

Specimen	How	Trap
Blood culture	2 sets 2 sites, fever spike, 20-30 mL adult, povidone + alcohol, before Ab	Single set = skin contaminant confusion
Urine	Midstream clean-catch, peri-wash, sterile wide-mouth, <2h or 4°C 24h	First-stream + mixed flora = contamination
Sputum	Early morning deep cough, rinse water only, not saliva	Saliva reported as normal flora — repeat

Speed Check: Urine hold if delay?

4°C up to 24h, never room temp overnight.

WOUND, CSF, STOOL TRANSPORT

Aspirate over swab, ice vs room

A+

TRANSPORT MATRIX

Specimen	Container / Temp
Wound pus aspirate	Sterile syringe + anaerobic vial, room temp urgent
CSF	3 tubes sterile, warm, never refrigerate (meningococcus dies), urgent
Stool O&P	Clean dry pot, no urine mixing, <2h; Cary-Blair for culture
Viral swab	VTM on ice, no formalin for PCR

FATAL TRAP: Refrigerating CSF. Kills meningococcus + Hib — keep warm, rush to lab.

NORCET HIT: CSF warm-urgent, urine cold-wait, virus ice-VTM.

Speed Check: CSF storage?

Room temp/warm, immediate — never fridge.

STERILIZATION MONITORING BUNDLE

Physical, chemical, biological

A**PCD + LOG**

- **Every cycle:** temp/pressure/time log + external tape + internal strip.
- **Daily:** Bowie-Dick air removal (pre-vacuum). **Weekly:** biological.
- **Recall:** positive biological → quarantine load, repeat, inform HICC.

Speed Check: Bowie-Dick tests?

Air removal in pre-vacuum autoclave.

OT STERILITY: FUMIGATION TO TRAFFIC

Count, traffic, temp-humidity

A

OT CARD

- **Traffic:** minimal, doors closed, 20-24°C, 30-60% humidity, positive pressure, HEPA.
- **Counts:** sponge/instrument count pre + before closure; sign board.
- **Between cases:** surface disinfection, air settling time.

Speed Check: OT temp-humidity?

20-24°C, 30-60% RH.

WHY STERILIZATION FAILS

Overload to wet packs

A+

FAILURE AUDIT

- Overloading, wrong temp-time, air pockets, wet packs, expired indicators, faulty gaskets.
- **Nurse action:** do NOT use doubtful pack — return to CSSD, document, take replacement.

Speed Check: Wet pack use?

Never — consider contaminated.

COMPLIANCE AUDIT IN WARD

Observe, feedback, bottles

A

AUDIT CUE

- **Direct observation** of 5 moments; compliance % = actions/opportunities x100.
- **Ensure:** bedside rub bottles filled, posters, monthly feedback.

Speed Check: Compliance formula?

Performed / opportunities x100.

STAFF SAFETY DURING COLLECTION

PPE + sharps + spill ready

A+

SAFETY FIRST

- Gloves + mask-eye for splash, no needle recapping, leak-proof bagging, hand hygiene after.
- **NSI:** wash, report, risk assess (preview Batch 9).

Speed Check: Bagging?

Double-bag leaking, biohazard label.

BATCH 5 BLITZ (18 IN 3 MIN)

Hands + sterilization + specimen

A+**18 Q&A**

#	Q	A
1	5th moment?	After surroundings
2	Rub time?	20-30 sec
3	Wash time?	40-60 sec
4	C.diff hands?	Soap wash
5	Spaulding critical?	Sterilize
6	Autoclave?	121-15-15
7	Autoclave spore?	Geobacillus
8	Dry heat?	160-1h oils/powders
9	Blood spill?	1% hypo
10	Scope HLD?	Glut 2% 20-45 min
11	CSF store?	Warm urgent
12	Blood sets?	2 sets 2 sites
13	Urine delay?	4°C 24h
14	Bowie-Dick?	Air removal
15	OT climate?	20-24°C 30-60%
16	Wet pack?	Reject
17	Missed areas?	Thumbs/fingertips
18	Next?	BMW + AMR

BMW 4 COLOURS AT A GLANCE

Segregate at bedside, never mix

A+

COLOUR MASTER

Core: Segregate at generation point into labelled colour bags. Tie when 3/4 full. Store <48 hrs. Verify against current BMW chart in your state before exam.

Colour	Goes in	Treatment
YELLOW	Infectious: dressings, tissues, cultures, diapers, bedding contaminated	Incineration / deep burial
RED	Contaminated recyclable: tubing, bottles, gloves, urine bags	Autoclave / chemical + recycle
WHITE (translucent)	Sharps: needles, blades, metal sharps	Puncture-proof box, autoclave/shred
BLUE	Glass + metal implants: vials, slides	Disinfection + recycle

Dynamic rule: BMW Rules 2016 + amendments. Colour shades/bag specs update — recheck hospital chart + pollution board poster before NORCET.

Speed Check: Dressing with pus goes to?

YELLOW bag — infectious, incineration.

YELLOW BAG: INFECTIOUS MASTER

What exactly qualifies

A+**YELLOW CARD**

- **In:** soiled dressings, cotton, placenta/tissues, cultures/stocks, diapers with blood, linen heavily soiled (as per local tie-up), expired/discarded medicines (cytotoxic separate black as per facility).
- **Bag:** non-chlorinated yellow, barcoded, tied 3/4, biohazard label.
- **Never in yellow:** sharps, glass, routine gloves, food waste, general trash.

FATAL TRAP: Dumping sharps + glass in yellow. Punctures bag, injures handler — sharps strictly white, glass blue.

NORCET HIT: Wet infectious = Yellow. Dry recyclable plastic = Red.

Speed Check: Culture plates after reading?

Yellow — autoclave/incinerate, never general bin.

RED BAG: RECYCLABLE PLASTIC

Tubes, bottles, gloves

A+**RED CARD**

- **In:** IV tubing, urine bags, gloves, plastic bottles, syringes WITHOUT needles (after hub-cut).
- **After:** autoclave/microwave/chemical then recycle — never incinerate chlorinated unnecessarily.
- **Hub-cut:** cut needle to white box immediately at bedside, syringe body to red.

Speed Check: IV set after removal?

Red bag — recyclable contaminated plastic.

WHITE SHARPS & BLUE GLASS

Puncture-proof + leak-proof

A+

SHARPS CARD

Box	Items	Rule
WHITE puncture-proof	Needles, blades, scalpels, metal sharps	No recapping, drop immediately, 3/4 seal, label SHARPS
BLUE box/bag	Glass vials, slides, metal implants	Disinfect, intact transport, recycle

- **Hub-cutter** at every injection tray; bleach in cutter box as per protocol.

FATAL TRAP: Recapping needle with two hands. Single-hand scoop only if must — best is immediate drop, no recap.

NORCET HIT: Sharp = White. Glass = Blue.

Speed Check: Box fill limit?

Seal at 3/4 full, never press down.

LABEL, STORE, TRANSPORT <48 HRS

Barcode to vehicle

A+

LOGISTICS

- **Label:** ward, date-time, colour, weight, barcode; biohazard + cytotoxic symbols where needed.
- **Store:** locked central room, ventilated, washable floor, <48 hrs, separate general waste route/time.
- **Transport:** closed trolley, no manual carrying long distance, spill kit + PPE with handler.
- **Records:** BMW register, annual report to board, accident report.

Speed Check: Max storage time?

48 hours untreated BMW.

GENERAL WASTE & FOOD LEFTOVERS

Green/black bins, no mix

A**NON-BMW**

- **General:** paper, wrappers, food peels → compost/landfill stream; never in yellow/red.
- **Mixing penalty:** whole bag becomes BMW if mixed — costs + infection.
- **Handler PPE:** heavy gloves, boots, apron, mask; immunize (HepB, tetanus).

Speed Check: Fruit peel in ward?

General waste — never yellow.

LAUNDRY: SLUICE TO STERILE

Bag, wash, iron, pack

A+**LAUNDRY FLOW**

- **Bag at bedside** colour-coded, no shaking/counting on bed; tie + label infected.
- **Sluice:** cold rinse → hot wash 70-80°C + detergent → bleach → dry → iron (heat kills) → pack.
- **Staff:** gloves + apron, hand hygiene, separate clean/dirty zones, no hugging linen.
- **Infected linen:** pre-soak hypochlorite as per local SOP, machine wash, sun dry where allowed.

FATAL TRAP: Shaking/counting linen in ward. Aerosolizes germs — bag, don't sort at bedside.

NORCET HIT: Hot wash + Iron = disinfection of linen.

Speed Check: Wash temp?

70-80°C with detergent.

WARD BMW AUDIT CHECKLIST

Bins, bags, needles

A

AUDIT CUE

- Bins with lids + colour liners + labels at generation points? Bags 3/4 tied + barcoded? Hub-cutter with bleach? Sharps box sealed on time? Handler PPE + HepB done? Register updated?

Speed Check: First fix for mixed bags?

Re-train segregation at bedside + reposition bins.

AMR: WHY DRUGS FAIL

MRSA, VRE, CRE, ESBL

A+

AMR BASE

Bug	Resists	Spread block
MRSA	All beta-lactams (mecA/PBP2a)	Contact + mupirocin + CHG bath
VRE	Vancomycin (vanA)	Contact + antibiotic review
CRE/ESBL	Carbapenems/cephalosporins	Contact + carbapenem guard
MDR-TB/XDR	Rifampicin ± second-line	Airborne + DOTS adherence

- **Causes:** overuse, wrong dose/duration, broad without culture, patient non-adherence, OTC sale.

Speed Check: Gene for MRSA?

mecA → PBP2a.

NURSE ROLE IN STEWARDSHIP

Culture first, review 48-72h

A+**STEWARD CARD**

- **Before Ab:** cultures + indication + allergy + renal/liver + pregnancy documented.
- **During:** right dose-time-route, de-escalate IV→oral on improvement, stop date reviewed 48-72h.
- **Teach:** полный course, no sharing, no OTC, red-flag diarrhoea (C.diff).
- **Guard:** restricted/high-end need approval + culture proof.

FATAL TRAP: Starting broad antibiotic without cultures then chasing reports. Culture FIRST or stewardship fails.

NORCET HIT: Right drug-bug-dose-duration-route = stewardship.

Speed Check: Review window?

48-72 hours de-escalation review.

ANTIBIOGRAM & RESTRICTED LIST

Read S-I-R + approval

A**S-I-R CARD**

- **S = sensitive**, I = intermediate (high dose/site), R = resistant. Treat by S narrowest.
- **Restricted:** carbapenems, linezolid, colistin — culture + consultant + HICC note.
- **Prophylaxis:** SSI single dose 60-min rule, no prolonged post-op cover.

Speed Check: Pick which drug?

Narrowest S, correct site penetration.

MDRO PATIENT CARE BUNDLE

Isolate + inform + clean

A+

MDRO BUNDLE

- Single/cohort, contact PPE, dedicated kit, hand hygiene, daily CHG bath (MRSA), environment bleach clean, flag chart, inform OT/dialysis.
- **Decolonize:** mupirocin nasal + CHG wash 5 days as ordered; repeat screen.

Speed Check: Mupirocin site?

Anterior nares.

BMW SCENARIOS SOLVED

Bin picks in seconds

A+

SCENARIO GRID

Item	Bin
Blood-soaked gauze	Yellow
Urine bag + tubing	Red
Used needle	White
Broken ampoule	Blue
Gloves after dressing	Red (if contaminated recyclable) per local chart

Speed Check: Placenta?

Yellow — human tissue.

AMR VIGNETTES SOLVED

Pick action, not just drug

A+**VIGNETTE GRID**

Vignette	Action
Fever + line + MRSA screen +	Contact + cultures + vancomycin per protocol + remove line if infected
Diarrhoea post-clinda	Suspect C.diff, contact + soap, stop clinda, test
UTI + ESBL	Contact, carbapenem only if sensitive + approved, catheter review

Speed Check: C.diff drug trigger classic?

Clindamycin (also fluoroquinolones).

CYTOTOXIC & EXPIRED DRUGS

Black bag + spill kit

A**HAZARD CARD**

- **Separate stream** (black/return to pharmacy as per facility): chemo vials, expired drugs — never yellow general.
- **Spill:** PPE + spill kit + closed area + incident report.

Speed Check: Chemo spill first?

PPE + contain + call trained team.

HANDLER & NURSE SAFETY IN BMW

Vaccinate + PPE + training

A

SAFETY CUE

- HepB + tetanus for handlers; heavy gloves + boots; annual health check; NSI protocol known.
- **Training:** induction + yearly, posters in local language, mock spill drill.

Speed Check: Handler vaccines?

HepB + tetanus.

REGISTERS & ANNUAL REPORT

Weight, barcode, accident

A

RECORD CARD

- Daily weight colour-wise, handover signature, vehicle manifest, accident + annual report to board.

Speed Check: Who signs handover?

Ward nurse + transporter + central store.

BATCH 6 BLITZ (18 IN 3 MIN)

BMW + AMR

A+**18 Q&A**

#	Q	A
1	Pus dressing?	Yellow
2	IV tubing?	Red
3	Needle?	White 3/4
4	Ampoule glass?	Blue
5	Store time?	48h
6	Laundry temp?	70-80°C
7	Shake linen?	Never
8	MRSA gene?	mecA
9	Review?	48-72h
10	Culture when?	Before Ab
11	S-I-R pick?	Narrowest S
12	Mupirocin where?	Nares
13	Placenta?	Yellow
14	C.diff trigger?	Clindamycin
15	Handler vaccine?	HepB+TT
16	Bag fill?	3/4 tie
17	Recap?	No, drop in white
18	Next?	Safety + IPSC

FALLS, RESTRAINTS & VULNERABLE CARE

Morse, least restraint, rails

A+**FALL + RESTRAINT**

Core: Falls = top reported incident. Restraint = last resort with order + monitoring q15-30 min.

- **Morse risks:** history fall, secondary diagnosis, ambulatory aid, IV, gait, mental status. High score = yellow band + low bed + rails + call bell + hourly round.
- **Prevent:** low bed + locked brakes + night light + non-slip + Review meds (sedatives) + accompany toilet.
- **Restraint:** physician order time-boxed, least restrictive, padding, circulation + skin + toileting checks, family consent + de-escalate fast.
- **Post-fall:** vitals + neuro + injury check, inform doctor/family, incident form, RCA, shift handover flag.

FATAL TRAP: Raising all 4 rails without order as routine. Considered restraint — needs assessment + order + monitoring.

NORCET HIT: Low bed + Bell + Brakes + Light = fall bundle.

Speed Check: First after fall?

Assess Airway-Breathing-Circulation + neuro + injury before moving.

TRANSFUSION: CHECK TO REACTION

Two-nurse check, vitals, stop rule

A+**BLOOD CARD**

- **Before:** consent, group-crossmatch tag, 2-nurse bedside check (name/ID/bag/no/expiry), vitals, IV 18-20G saline, filter set.
- **Start slow:** first 15 min 2 mL/min bedside stay; vitals 15 min → 30 min → hourly → end.
- **Reaction:** fever/chills/flank/red urine/wheeze → STOP, saline, notify, return bag + sample, document.
- **Storage:** issue → transfuse within 30 min / 4 hrs complete; never fridge ward + never warm in hot water.

FATAL TRAP: Adding drugs to blood bag. Hemolysis + clots — saline only, separate line.

NORCET HIT: Saline only + Slow first 15 + Stay bedside.

Speed Check: Fluid with blood?

Normal saline only.

IV COMPLICATIONS & LINE SAFETY

Phlebitis to infiltration

A+

IV CARD

Complication	Signs	Action
Infiltration	Cool, pale, swelling, slow flow	Stop, elevate, warm/cold per drug, restart proximal
Extravasation (vesicant)	Blister/necrosis (chemo, dopamine)	Stop, aspirate, antidote per protocol, plastics review
Phlebitis	Red, warm, cord, pain (VIP score)	Remove, warm compress, new site, document VIP
Fluid overload	Crackles, JVD, high BP	Slow/stop, sit up, O2, notify, I/O strict

Speed Check: VIP score checks?

Redness, pain, swelling, palpable cord.

DVT PROPHYLAXIS & SAFE TRANSPORT

Stockings, heparin, handover

A+

DVT + SBAR

- **DVT:** early mobilize, leg exercises, stockings/SCD, hydration, heparin as ordered; Homans NOT reliable — Doppler.
- **Transport:** SBAR handover, O2 + suction + monitor for critical, rails up, brakes, belongings + file.
- **Med rec:** admission vs discharge list match; high-alert double-check.

Speed Check: Best DVT prevention?

Early ambulation + mechanical + pharma as ordered.

SAFE SURGERY CHECKLIST

Sign in, time out, sign out

A+**WHO CHECKLIST**

- **Sign in:** identity + consent + site mark + anesthesia + allergies + blood ready.
- **Time out:** all pause, surgeon-nurse-anesthesia confirm patient-site-procedure-antibiotics + images.
- **Sign out:** counts, specimens labelled, recovery plan, handover.
- **Site mark:** surgeon marks with patient awake, visible after drape.

FATAL TRAP: Skipping time out for hurry. Wrong-site surgery never event — pause mandatory.

NORCET HIT: Mark awake + Pause together + Count aloud.

Speed Check: Who marks site?

Operating surgeon with patient awake.

DOCUMENTATION & HANDOVER SAFETY

SBAR, read-back, no verbal orders

A

COMM CARD

- **SBAR:** Situation-Background-Assessment-Recommendation for escalation.
- **Read-back:** repeat verbal/telephone orders + critical values; written ASAP + countersign.
- **Do NOT use:** banned abbreviations (U, IU, QD, trailing zero); 24-hr time; legible + timed + signed.

Speed Check: Critical value action?

Read-back + act + document time + inform doctor.

INCIDENT CAPTURE & REPORT WRITING

What, when, no blame

A**REPORT CARD**

- **Capture:** fall, med error, NSI, pressure injury, HAI, near miss — immediate patient safety first.
- **Write:** facts only (date-time-place-person-event-action), no opinions/blame, vitals + witnesses, inform in-charge.
- **Never:** hide, overwrite, or blame in file; incident book ≠ patient file accusation.

Speed Check: First in incident?

Secure patient safety, then report.

RCA (5 WHYS) & CAPA LOOP

System fix, not person blame

A**RCA TOOL**

- **RCA steps:** gather facts → timeline → 5 whys/fishbone → root causes (system) → CAPA → follow-up audit.
- **Corrective:** fix now (re-train, replace bottle). **Preventive:** stop repeat (checklist, alarm, stock).
- **Example:** fall → why rails down → why no assessment → why no form → fix form + training + audit.

Speed Check: RCA asks?

5 whys to system root, not who.

PRESSURE INJURY PREVENTION

Braden, 2-hour turn, heel off

A+

BRADEN CARD

- **Braden risks:** sensory, moisture, activity, mobility, nutrition, friction. Low score = high risk.
- **Prevent:** 2-hr turn, 30° lateral, heel float, air mattress, dry + barrier cream, protein + fluids.
- **Stage fast:** 1 non-blanch red → 2 blister → 3 fat visible → 4 muscle/bone → unstageable/suspected deep.

Speed Check: Turn interval?

Every 2 hours + skin check.

UNIT VIII RAPID TABLES (8 PAGES CONDENSED)

Vulnerable, iatrogenic, audit cues

A+

REVISION GRID

Risk	Prevent	Audit
Elderly/confused	Low bed, bell, attendant, no sedative dusk	Fall rate, restraint hours
Lines/drains	Secure, label, daily need, sterile handling	Accidental removal count
Medication	5 rights + high-alert double + allergy band	Error + near-miss rate
Communication	SBAR + read-back + no banned short forms	Handover compliance
HAI	Bundles + hand + isolation	Device-day rates
Documentation	Timed, signed, factual	Chart audit

- Pages 10-17 expand each row with vignette + trap + speed check in PDF (full exposition kept concise for print).
- **iatrogenic watch:** contrast allergy ask, vallabel, allergy band, surgical count, specimen label bedside.

Speed Check: Best fall metric?

Falls per 1000 patient-days + injury rate.

BATCH 7 BLITZ

Safety + RCA

A+

Q&A

#	Q	A
1	Fall bundle?	Low bed-bell-brakes-light
2	Blood with?	Saline only
3	Transfusion first 15?	Slow + stay
4	Phlebitis scale?	VIP
5	DVT best?	Ambulate
6	Time out when?	Before incision pause
7	SBAR for?	Escalation
8	Read-back for?	Verbal/critical
9	RCA asks?	5 whys system
10	Turn time?	2 hr

IPSG 1-2: IDENTIFY + COMMUNICATE

Two identifiers + SBAR + read-back

A+

IPSG CARD

Core: IPGS 1 = correct patient (2 identifiers, never room/bed). IPGS 2 = effective communication (SBAR + read-back).

- **2 identifiers:** name + ID/DOB/barcode; band check before drug, sample, procedure, feed.
- **SBAR escalation,** read-back verbal + critical values, no banned abbreviations.

FATAL TRAP: Room-number identification. Always 2 person-identifiers.

NORCET HIT: Name + ID, SBAR + Read-back.

Speed Check: Identifiers?

Name + hospital ID/DOB.

IPSC 3-4: HIGH-ALERT DRUGS + SAFE SURGERY

Double-check + checklist

A+

HAM CARD

- **High-alert (HAM):** heparin, insulin, chemo, KCl — double-check, locked store, red label, daily review.
- **LASA:** separate storage, tall-man lettering, read-back.
- **Surgery:** mark awake + time out + counts + specimen label bedside.

Speed Check: KCl storage?

Locked, diluted only, never IV push.

IPSG 5-6: HAI + FALLS + ALARMS

Bundles + Morse + alarm fatigue

A+

IPSG FINISHER

- **HAI:** hand + bundles + isolation (Batch 4). **Falls:** Morse + low bed + bell.
- **Alarms:** set limits per patient, respond, no silencing without check, daily electrode change.

Speed Check: Alarm fatigue fix?

Individualize limits + respond every alarm.

5S, RISK ASSESSMENT & AUDITS

Sort to Sustain + temp-humidity

A+

5S CARD

- **5S:** Sort-Set-Shine-Standardize-Sustain. Crash cart checked daily, expiry FIFO.
- **Environment:** OT 20-24°C 30-60%, ward ventilation, lux, noise control.
- **Audit:** hand, bundle, BMW, med, fall — run charts + feedback.

Speed Check: 5S first S?

Sort — remove unneeded.

FIRE: RACE + PASS + EXTINGUISHER

Rescue-Alarm-Confine-Extinguish

A+

FIRE CARD

Class	Fuel	Agent
A	Wood/paper/linen	Water/ABC powder
B	Oil/petrol	Foam/CO2, never water jet
C/Electrical	Panel, wires	CO2/dry powder, power off first

- **RACE:** Rescue-Alarm-Confine-Extinguish/Evacuate. **PASS:** Pull-Aim-Squeeze-Sweep.
- **Nurse:** close doors/windows, O2 off, lift horizontal evacuation, headcount, no lift use.

FATAL TRAP: Water on electrical/oil fire. Shock + spread — CO2/powder only.

NORCET HIT: RACE to rescue, PASS to spray.

Speed Check: Electrical fire agent?

CO2 after power off.

HAZMAT, MSDS & CHEMICAL SPILL

Read SDS before use

A+**SDS CARD**

- **SDS 16 sections:** hazards, PPE, first aid, storage, disposal. Read before opening new chemical.
- **Spill:** alert, PPE, contain, absorb/hypo per agent, ventilate, report.
- **Mercury:** never vacuum; sulfur powder + collect + sealed bottle.

Speed Check: Mercury spill tool?

Sulfur powder + cardboard, no vacuum.

RADIATION ALARA & LASER SAFETY

Time-Distance-Shielding

A+

ALARA CARD

- **ALARA:** As Low As Reasonably Achievable — least time, max distance, lead shield.
- **Nurse:** lead apron + thyroid collar + dosimeter (TLD), pregnant staff declare + rotate out, mobile X-ray announce + step back 2 m.
- **Laser:** goggles wavelength-specific, no reflective instruments, wet drapes for airway laser, smoke evacuator.

Speed Check: ALARA stands for?

As Low As Reasonably Achievable.

EMERGENCY CODES: BLUE, RED, PINK

Call, crash cart, crowd control

A+**CODE CARD**

Code	Means (verify local)	Nurse first
Blue	Cardiac arrest	Call + start CPR + defib pads + crash cart
Red	Fire	RACE + alarm + confine
Pink/White	Infant abduction / violence (local varies)	Lock exits + headcount + security

Codes vary by hospital — memorize YOUR hospital code chart; exam asks generic RACE/CPR first actions.

Speed Check: First in Code Blue?

Call for help + start high-quality CPR.

ELECTRICAL & EQUIPMENT SAFETY

3-pin, no overload, calibration

A

EQUIPMENT CARD

- 3-pin grounded plugs, no multi-overload, dry hands for switches, report sparking, biomedical stickers valid.
- **Crash cart:** sealed, daily check, defib test, drug expiry FIFO.

Speed Check: Wet hands + plug?

Never — dry + insulated.

DISASTER TRIAGE: START COLOURS

Red-Yellow-Green-Black

A+

TRIAGE CARD

Colour	Means	Example
Red immediate	Airway/bleeding, salvageable	Chest wound, arterial bleed
Yellow delayed	Stable, can wait	Closed fracture
Green minor	Walking wounded	Abrasions
Black expectant/dead	Unsalvageable/deceased	Massive head crush

- **Nurse role:** tag, bleeding control, airway, count, reunite families, document.

Speed Check: First triage category?

Red immediate salvageable.

SAFETY DRILLS BANK (7 SCENARIOS)

Fire, spill, needle, power, oxygen, fall, code

A+**DRILL BANK**

- 11 Fire in linen room → RACE + close + O2 off + evacuate horizontal.
- 12 Formalin spill eye → irrigate 15-30 min + eye OPD.
- 13 Power cut in ICU → manual Ambu + call engineer + triage vents.
- 14 O2 leak smell → no spark + close valve + ventilate + evacuate.
- 15 Fall with head injury → C-spine + neuro + CT per order.
- 16 Wrong-drug near miss → hold + vitals + inform + incident + RCA.
- 17 Alarm muted death → never silence without viewing patient.

Speed Check: O2 leak + spark?

No switch/flame — close valve, ventilate.

BATCH 8 BLITZ

IPSG + protocols

A+

Q&A

#	Q	A
1	Identifiers?	Name+ID
2	KCI rule?	Never push, locked
3	5S first?	Sort
4	RACE?	Rescue-Alarm-Confine-Extinguish
5	PASS?	Pull-Aim-Squeeze-Sweep
6	Electric fire?	CO2
7	ALARA?	Time-distance-shield
8	Code Blue first?	Call+CPR
9	Triage red?	Immediate
10	O2 leak?	No spark, close valve

NSI FIRST AID: WASH, REPORT, ASSESS

Professional Nursing Coach • NUR-07 Employee Safety + Mega Review

A+

BATCH 9

Core: Wash under running water + soap (no squeezing/sucking), antiseptic, report within 30 min to CMO/ART centre, fill NSI form.

Step	Action
1 Wash	Soap-water, no scrub/squeeze; eye splash irrigate 15 min
2 Report	In-charge + infection control + ART; time-source noted
3 Assess	Source HBsAg/HCV/HIV + exposed vaccine/Ab status
4 PEP	HIV <72h (best 2h) 28 days; HBV vaccine/HBIG; HCV follow-up

FATAL TRAP: Do NOT squeeze wound to bleed — increases trauma. Wash, don't suck.**NORCET HIT:** Washed-Reported-Assessed-PEP in 2 hours best.

HIV PEP: 72H WINDOW, 28 DAYS

Professional Nursing Coach • NUR-07 Employee Safety + Mega Review

A+

BATCH 9

Core: Risk: deep hollow-bore + visible blood + advanced source = high. Start TLD-based PEP per current NACO chart; recheck guideline box before exam.

Risk	Example
High	Deep stick, hollow needle, blood, high viral load
Low	Solid needle, superficial, no blood
Mucosal	Splash to eye/mouth — irrigate, assess volume

FATAL TRAP: Delay beyond 72h = PEP not recommended. Start first dose ASAP, labs parallel.

NORCET HIT: 2-hour start saves life; 28-day complete.

HBV PEP: VACCINE + HBIG MATRIX

Professional Nursing Coach • NUR-07 Employee Safety + Mega Review

A+

BATCH 9

Core: Exposed unvaccinated + HBsAg+ source = vaccine + HBIG. Prior vaccinated with anti-HBs ≥ 10 = protected.

Exposed	Source +	Action
Unvaccinated	HBsAg+	HBV vaccine series + HBIG IM
Vaccinated responder	Any	No action, reassure
Non-responder	HBsAg+	HBIG + revaccinate

FATAL TRAP: HBIG within 24h best (up to 7 days). Document lot + site.**NORCET HIT:** Anti-HBs ≥ 10 mIU/mL = protected.

HCV + TETANUS + RABIES LINKS

Professional Nursing Coach • NUR-07 Employee Safety + Mega Review

A+

BATCH 9

Core: HCV: no PEP vaccine — baseline + 6-week/3-month/6-month RNA/Ab follow-up. Tetanus: dirty wound + unknown TT = TIG + TT.
Animal bite: wash + RIG + ARV.

Exposure	Follow-up
HCV stick	0, 6wk, 3mo, 6mo testing; counsel, no donation
Dirty wound	TT/TIG per immunization history
Dog bite	Category II/III wash + vaccine ± RIG

FATAL TRAP: Reassure but follow-up complete — no early all-clear.

NORCET HIT: Test-Test-Test till 6 months for HCV.

HEALTHCARE WORKER VACCINES

Professional Nursing Coach • NUR-07 Employee Safety + Mega Review

A+

BATCH 9

Core: HepB 0-1-6 + anti-HBs check, TT/Td boosters, annual flu, MR/varicella if non-immune, COVID as per current policy.

Vaccine	Schedule cue
HepB	0-1-6 months, check titre 1-2 mo after
TT/Td	Booster 10 years / dirty wound 5-year rule
Flu/MR	Annual / outbreak non-immune staff

FATAL TRAP: Post records in staff health file — needed for posting + NSI claims.**NORCET HIT:** File your own card like patient card.

ANNUAL HEALTH + RADIATION + FALLS (STAFF)

Professional Nursing Coach • NUR-07 Employee Safety + Mega Review

A+

BATCH 9

Core: Annual check, TB screen, fitness for N95, pregnant rotation, fall-safe wards, violence code white.

Check	Includes
Annual	CBC, LFT, chest, vision, HepB titre, TB symptom screen
Radiation	TLD badge monthly, declare pregnancy early
Violence	De-escalate, panic button, incident + counselling

FATAL TRAP: Never ignore chronic cough in staff — TB screen + duty modification.**NORCET HIT:** Healthy nurse = safe patient.

ERGONOMICS: BACK, SHIFT, BURNOUT

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BATCH 9

Core: Body mechanics (bend knees, wide base), shift sleep hygiene, burnout watch, sharps mindfulness.

Risk	Prevent
Back injury	Raise bed, 2-person lift, slide sheets
Needle fatigue	No rushing last sample, tray organized
Burnout	Roster equity, breaks, peer support

FATAL TRAP: Sleepy night-shift stick = commonest NSI story. Rest + focus.**NORCET HIT:** Lift with legs, not back.

NSI SCENARIOS SOLVED

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BATCH 9

Core: Deep IM needle from HIV+ → wash + start PEP 2h + labs. Superficial suture needle, source negative → wash + observe + document.

Vignette	Answer
Eye splash blood	Irrigate 15 min + risk assess + PEP if high
Gloved needle, no blood	Low risk, wash + document + follow-up
HBsAg+ stick, unvaccinated	Vaccine + HBIG + follow-up

FATAL TRAP: Document source consent + counselling — medico-legal must.

NORCET HIT: Scenario decides PEP, not fear.

BMW-NSI-LAUNDRY LINK

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BATCH 9

Core: Segregate sharps white + laundry bagged + spill hypo = NSI down. Audit monthly.

Link	Action
White box 3/4	Seal on time, no pressing
Linen	No sharps in linen — scan before bagging
Spill	PPE + hypo + report

FATAL TRAP: Most NSI from overfilled boxes + linen-hidden sharps.**NORCET HIT:** Box + Bag + Glove = no stick.

MEGA BLITZ 1: MICRO TO MEDIA

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BATCH 9

Core: Leeuwenhoek-Gram-ZN-KOH-LJ-TCBS-SDA streak-anaerobe.

Q	A
Gram decolorizer	Acetone-alcohol 10-20 sec
ZN decolorizer	20% H ₂ SO ₄
Capsule	India ink halo
LJ for	TB 2-8 wks
TCBS yellow	Vibrio

FATAL TRAP: Batch 1-2 core in 60 seconds.**NORCET HIT:** Stain decides bug.

MEGA BLITZ 2: IMMUNITY TO VACCINES

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BATCH 9

Core: IgG placenta, IgM acute, IgA milk, IgE allergy; adrenaline thigh; Mantoux 48-72h; BCG left-ID; cold 2-8C.

Q	A
Placental	IgG
First	IgM
Anaphylaxis	IM adrenaline 0.5 thigh
Mantoux	48-72h induration
VVM discard	Inner darker

FATAL TRAP: Batch 3 in 60 seconds.**NORCET HIT:** G-M-A-E + thigh.

MEGA BLITZ 3: HAI TO PPE

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BATCH 9

Core: 48h HAI; CAUTI closed-low-review; CLABSI scrub 15; SSI clip; VAP head 30-45; droplet 3ft surgical; airborne N95 negative.

Q	A
HAI cutoff	48h
Scrub hub	15 sec
Cuff	20-30 cmH ₂ O
Droplet	3ft surgical
Airborne trio	M-T-C

FATAL TRAP: Batch 4 in 60 seconds.**NORCET HIT:** Device-days x1000.

MEGA BLITZ 4: HANDS TO SPECIMEN

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BATCH 9

Core: Rub 20-30, wash 40-60; Spaulding critical-sterilize; autoclave 121-15-15; blood spill 1% hypo; CSF warm urgent; 2 sets cultures.

Q	A
Rub/wash	20-30 / 40-60
Autoclave	121-15-15 Geobacillus
Blood spill	1% hypo
CSF	Warm urgent
C.diff hands	Soap wash

FATAL TRAP: Batch 5 in 60 seconds.**NORCET HIT:** Critical = sterilize.

MEGA BLITZ 5: BMW TO AMR

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BATCH 9

Core: Yellow infectious, red plastic, white sharps, blue glass; 48h store; mecA MRSA; culture first; 48-72h review.

Q	A
Pus dressing	Yellow
Needle	White 3/4
Glass	Blue
MRSA gene	mecA
Review	48-72h

FATAL TRAP: Batch 6 in 60 seconds.**NORCET HIT:** Segregate at bedside.

MEGA BLITZ 6: SAFETY TO IPSG

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BATCH 9

Core: Fall low-bed-bell; blood saline-slow; RACE-PASS; ALARA; triage red immediate; name+ID; KCI locked.

Q	A
RACE	Rescue-Alarm-Confine-Extinguish
PASS	Pull-Aim-Squeeze-Sweep
ALARA	Time-distance-shield
Triage red	Immediate
Identifiers	Name+ID

FATAL TRAP: Batch 7-8 in 60 seconds.**NORCET HIT:** Pause + check + count.

NSI FINAL DRILL + HANDOVER

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BATCH 9

Core: Every NSI: wash-report-assess-PEP-followup-document-counsel. Keep staff cards updated. Next subject: choose NUR-06 or NN series.

Handover	Done
Batches 1-9	Sec A I-IV + Sec B I-XI complete per blueprint
Files	9 PDFs in NUR-07 folder
Next	Merge to compendium now

FATAL TRAP: Zero NSI hiding — report every prick.**NORCET HIT:** Compendium next.